

Optimal control for an SIR model with limited hospitalised patients

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ABSTRACT

This paper analyses the optimal control of infectious disease propagation using a classic susceptible–infected–recovered (SIR) model characterised by permanent immunity and the absence of available vaccines. The control is performed over a time-dependent mean reproduction number, in order to minimise the cumulative number of ever-infected individuals (recovered), under different constraints. We consider constraints on non-pharmaceutical interventions ranging from partial lockdown to non-intervention, as well as the social and economic costs associated with such interventions, and the capacity limitations of intensive care units that limits the number of infected individuals to a maximum allowed value. We rigorously derive an optimal quarantine strategy based on necessary optimality conditions. The obtained optimal strategy is of a boundary-bang type, comprising three phases: an initial phase with no intervention, a second phase maintaining the infected population at its maximum possible value, and a final phase of partial lockdown applied over a single interval. The optimal policy is further refined by optimising the transition times between these phases. We show that these results are in excellent agreement with the numerical solution of the problem.

1. Introduction

The outbreak of infectious diseases has always been a challenge to public health systems and society at large. Among the mathematical models used to understand and manage the spread of such diseases, the Susceptible–Infected–Recovered (SIR) model introduced by Kermack and McKendrick [1] has emerged as a fundamental framework. The SIR model provides valuable insights into the dynamics of epidemics, allowing us to quantify the impact of various control measures on disease transmission.

For many disease outbreaks when a vaccine is still not available, such as for Covid-19, different countries around the world had to take a series of unprecedented decisions as imposing non-pharmaceutical interventions like social distance, quarantines and total lock-downs as the most effective tools to mitigate the spread of the disease. Although these kinds of measures are helpful in reducing the virus transmission and giving time to health systems to adapt, they could be extremely stressful in terms of economic and social costs and, in longer periods, tend to have less compliance with the population.

In the context of managing infectious disease outbreaks, one critical concern is the pressure placed on healthcare systems, particularly

the capacity of hospitals to provide care to those in need. During epidemics, the demand for hospitalisation can often exceed available resources, leading to a higher mortality rate and a significant burden on healthcare workers.

We consider the classical SIR model widely used in epidemiology [2,3]. As it is usual in these models, births and deaths are neglected, and we assume that people who have recovered develop immunity and, therefore, would not be able to get infected nor infect others.

To account for quarantine measures, we consider a time-dependent reproduction number and we assume that the intervention will occur in the preset period of time $[0, T]$, as proposed by Greenhalgh [4] and more recently by Ketcheson [5], Bliman et al. [6] and Balderrama et al. [7], since it is unrealistic that interventions can be sustained indefinitely.

As mentioned in [6] (see references therein) there are other methods to control disease outbreaks in the framework of the SIR: pharmaceutical interventions as vaccination or treatment; or the screening and quarantining of infected. Also different modelling frameworks have been considered such as SEIR (with the addition of the Exposed compartment), SIRQ (with the addition of the Quarantine compartment)

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or SIRH (with the addition of the Hospitalised compartment) among others.

In this article we address the development and analysis of an optimal control framework that seeks to manage an epidemic by forcing social distancing. Specifically, the control aims to minimise the **final size of epidemic** while keeping the number of infections below a predetermined threshold and within the restrictions of the maximum social and economic cost that the policy maker can bear.

By incorporating constraints on the social and economic costs and the number of patients that can be admitted to hospitals, we aim to provide a realistic and actionable approach to epidemic management. In this regard, the maximum **social and economic cost** that the policy maker can bear is given by an L^1 -type restriction (i.e., an integral restriction) on the control, see the integral inequality (2). The limitation imposed by **hospital resources** is represented by a peak prevalence constraint on the number of infected individuals at any given time (i.e., a maximum allowable number of infections) as detailed in inequality (3). This is a running state constraint that considers the hospitalised patients as a fraction of the total infected population.

This research contributes to the ongoing effort to enhance our understanding of disease dynamics and offers practical strategies for policy makers and healthcare professionals to effectively combat infectious disease outbreaks while operating within real-world social, economic and healthcare constraints. The results shown in the present article conceptually reflect strategies that were implemented in real-time during the COVID-19 pandemic. For instance, in [8] the authors optimised the timing of toggling between mitigation stages via triggers that monitor daily new hospital admissions, while accounting for hospitalisations and minimising the duration of strict mitigation measures.

Recently, many works have appeared dealing with these and related issues. The problem of maximising the final number of susceptible individuals was studied by Ketcheson [5] in a prescribed intervention interval of time $[0, T]$. However, in that work it is assumed that the total lockdown corresponds to a zero reproduction number, something that is impossible to achieve in the real world. Moreover, it is assumed that the total lockdown can last during the whole intervention, which seems to be impracticable. Later, in [6] the authors considered the same optimal control problem with a partial lockdown. In these two articles it is proved that the optimal control is of bang–bang type consisting of a first phase of no action where no intervention measures are taken and a second phase where the maximum possible lockdown is applied until the end of the intervention time T . In [7] the same problem is studied, broadening the previous results by adding an L^1 -type restriction on the control accounting for a restriction on the social and economic cost. As a consequence, in this case the optimal control can have three phases: a first phase of no action and a second phase of partial lockdown during the maximum possible time that the L^1 -type restriction allows. In the present article, we extend the theoretical results obtained in [7] by adding the peak prevalence constraint for all time.

Dealing with running state constraints in optimal control theory adds a layer of complexity to our approach. These constraints reflect the dynamic and evolving nature of healthcare capacity, making it essential to adapt control strategies in real-time to ensure the efficient allocation of resources while minimising the disease's spread.

Several authors have analysed the problem of controlling an epidemic reducing transmission through lockdown policies, while keeping the infected below a certain threshold to prevent healthcare system saturation. Among them, Nenchev [9] considered an SIRQ-type model with a Mayer-type cost function. On the other hand, Avram [10], Sereno [11], and Miclo [12] studied this issue using an SIR-type model and an L^1 -type cost function for the control. Lastly, Morris [13] analysed an SIR-type model with a peak prevalence-type cost function and in [14] we studied the optimal final size with no L^1 -type restriction. In all these articles the optimal control obtained is of boundary-bang type.

In this article we employ a cost function different from those used in the previously mentioned works. Specifically, we use a final size of epidemic cost function with the aforementioned restrictions. Our goal is to characterise the optimal control for the problem of maximising the final size of the susceptibles with an L^1 -type restriction on the control and a running state constraint for the infected. Using an optimal control approach we show that the optimal strategy is a single *bang–bang* or *boundary-bang* type with a boundary control that keeps the infected in its maximum possible value followed by a partial lockdown applied in a single interval of time. Moreover, we characterise the time to start and finish each of these phases of the intervention.

In the following sections we shall elucidate the key parameters governing the behaviour of the considered SIR model, and present the optimal control strategies that can help mitigate the impact of epidemics under limited hospital capacity with real-time patient admission constraints and limited social and economic cost. Through rigorous analysis and simulation, we will show the potential of our approach to inform decision-making processes during public health crises.

The paper is organised as follows. We start by describing the optimal control model and preliminaries results in Section 2. In Section 3 we use Pontryagin's Maximum Principle (PMP) and show that only bang–bang or boundary-bang controls are admitted in a particular sequence to solve the corresponding optimal control problem. To obtain the specific optimal policy, the switching times between the different phases are determined in an induced optimisation problem in Section 4. In Section 5 we perform a numerical test of the results by finding approximate numerical solutions of the control problem and comparing them with the rigorous solution summarised in Theorem 4.1. Finally, in Section 6 we give a summary of the main findings and discuss future work.

2. Problem statement and preliminaries

The basic SIR compartmental model for infectious diseases divides a population into three homogeneous groups: Susceptible (S), Infected (I) and Recovered or Removed (R). The proportions of susceptible, infected, and recovered individuals at time t are denoted $x(t)$, $y(t)$ and $z(t)$, respectively. Since the total population is constant, we have that $z(t) = 1 - x(t) - y(t)$.

It is assumed that each infected individual comes into contact with an average of β random individuals per unit of time and can only infect those who are susceptible, leading to the transition $S \rightarrow I$ and generating new infections at an average rate of $\beta x(t)$. Additionally, each infected individual recovers at a rate γ , corresponding to the $I \rightarrow R$ transition. The infection rate β and recovery rate γ are related to the basic reproduction number σ , defined as $\sigma = \beta/\gamma$, which represents the average number of infections caused by a single infected individual in a completely susceptible population ($x = 1$) during its average infectious period of $1/\gamma$.

We address the challenge of managing the spread of an epidemic in the absence of vaccination and with limited hospital resources. The primary means of control available are non-pharmaceutical interventions (NPIs), represented by a time-dependent reproduction number, $\sigma(t)$, that can be adjusted within the range $[\sigma_s, \sigma_f]$. Here, σ_s signifies stricter NPIs (referred to as “strict” quarantine), while σ_f indicates a state without restrictions “free” from restrictions). The intervention is applied over a finite period $[0, T]$, where T represents the duration of the intervention. Following this period, restrictions are lifted, allowing the disease to spread freely with a constant $\sigma(t) = \sigma_f$ for all $t > T$. The parameter $\sigma(t)$ is conceptualised as encapsulating political measures like social distancing, business and institution lockdowns. Then the system evolves according to the following set of coupled nonlinear ordinary differential equations:

$$x'(t) = -\gamma \sigma(t) x(t) y(t), \quad (1a)$$

$$y'(t) = \gamma \sigma(t) x(t) y(t) - \gamma y(t), \quad (1b)$$

with $(x(0), y(0)) \in D = \{(x_0, y_0) : x_0 > 0, y_0 > 0, x_0 + y_0 \leq 1\}$, $\sigma(t) \in [\sigma_s, \sigma_f]$ for $t \in [0, T]$ and $\sigma(t) = \sigma_f$ for $t > T$, where $0 \leq \sigma_s < \sigma_f$. Note that the region D is forward-invariant and there exists a unique solution for all time [15].

Given that stricter NPIs lead to higher social and economic costs, we also assume that during the intervention period $[0, T]$, an extremely restrictive NPIs (corresponding to $\sigma = \sigma_s$) for an extended duration is deemed impractical. We impose the integral restriction:

$$\int_0^T \sigma(t) dt \geq \sigma_s \tau + \sigma_f (T - \tau), \quad (2)$$

with $\tau < T$ given. If σ is a control that takes the value σ_s for a period of time $\bar{\tau} > \tau$ we would have that $\int_0^T \sigma(t) dt \leq \sigma_s \bar{\tau} + \sigma_f (T - \bar{\tau}) < \sigma_s \tau + \sigma_f (T - \tau)$ contradicting inequality (2). Thus, the strict quarantine lasts at most τ time (see also inequality (42)). Note also that since long and strict NPIs have a social and economic negative impact, the lower bound on this restriction can be seen as an upper bound on the economic cost the policymaker can bear.

In addition, to prevent an overload of treatment resources and considering that a fraction of infected individuals will need to be hospitalised, we impose the constraint (running state constraint)

$$y(t) \leq K \text{ for all } t \in [0, T] \quad (3)$$

where $K > 0$ is a given constant.

After the intervention at time T , we calculate

$$x_\infty(x(T), y(T), \sigma_f) = \lim_{t \rightarrow \infty} x(t),$$

where $(x(t), y(t))$ is the solution of the system with initial condition $(x(T), y(T))$ and constant reproduction number $\sigma(t) \equiv \sigma_f$ for $t > T$.

Our objective is to determine the optimal control for the SIR model described above to minimise the overall impact of the pandemic in terms of the total number of infections. We aim to find an optimal $\sigma(t)$ in the family of piecewise continuous functions that minimises the total damage, considering the long-term behaviour of the system.

As it is known, for $y(0) > 0$, $\gamma \geq 0$ and $\sigma \geq 0$ the fraction of infected $y(t) > 0$ for any time $t \geq 0$ and approaches zero asymptotically, i.e., $y_\infty \equiv \lim_{t \rightarrow \infty} y(t) = 0$. The fraction of susceptible individuals $x(t)$ is strictly decreasing, and $x_\infty \equiv \lim_{t \rightarrow \infty} x(t) \in (0, 1/\sigma)$ (see Theorem 2.1 in [15]). Therefore, the state of the system in the long time limit consists only of susceptible and recovered individuals, $x_\infty + z_\infty = 1$, where $z_\infty \equiv \lim_{t \rightarrow \infty} z(t)$.

In order to find the optimal $\sigma(t)$ it proves convenient to work with the fraction of susceptible individuals in the long term x_∞ instead. Then, given that minimising z_∞ is equivalent to maximising x_∞ , we define the functional

$$J(x, y, \sigma) := x_\infty(x(T), y(T), \sigma_f). \quad (4)$$

Additionally, a new state variable $v(t) = \int_0^t \sigma(s) ds$ is introduced, enabling the study of the optimal control model

$$\max J(x, y, v, \sigma) := x_\infty(x(T), y(T), \sigma_f) \quad (5a)$$

$$\text{s.t. } x'(t) = -\gamma \sigma(t) x(t) y(t), \quad x(0) = x_0, \quad t \in [0, T], \quad (5b)$$

$$y'(t) = \gamma \sigma(t) x(t) y(t) - \gamma y(t), \quad y(0) = y_0, \quad t \in [0, T], \quad (5c)$$

$$v'(t) = \sigma(t), \quad v(0) = 0, \quad t \in [0, T], \quad (5d)$$

$$\sigma(t) \in [\sigma_s, \sigma_f], \quad \text{a.e. } t \in [0, T] \quad (5e)$$

$$v(T) \geq \sigma_s \tau + \sigma_f (T - \tau) \quad (5f)$$

$$y(t) \leq K, \quad t \in [0, T]. \quad (5g)$$

The optimal control problem (5) includes a pure running state constraint (5g), which is more challenging to address because it does not explicitly depend on the control variable σ . Therefore, it is useful to differentiate this constraint with respect to time t repeatedly until

the control variable appears (see [16]). To apply this approach, we introduce some classical definitions for optimal control problems with running state constraints, as outlined in [17]. These definitions can be skipped here and revisited when necessary. They will be employed mainly in Section 3.

An extremal arc of (5a)–(5f) is called an **unconstrained extremal** whereas an extremal arc of (5a)–(5g) is called a **constrained extremal**. For given K the state constraint (5g) is called **active** if the optimal unconstrained trajectory violates (5g).

By definition, the **order** p of an inequality pure state constraint is the first derivative of the restriction, $h(x(t), y(t), v(t)) = y(t) - K \leq 0$, containing the control σ explicitly. Since the control variable σ appears in the first total time derivative of $h(x(t), y(t), v(t))$

$$\frac{dh(x(t), y(t), v(t))}{dt} = y'(t) = \gamma y(t)(\sigma(t)x(t) - 1), \quad (6)$$

the state constraint is of order $p = 1$.

A subarc of the trajectory $(x(t), y(t), v(t))$ for which $h(x(t), y(t), v(t)) < K$ is called an **interior arc**; a **boundary arc** is a subarc of the trajectory where $h(x(t), y(t), v(t)) = K$ for $a \leq t \leq b$ with $0 < a < b < T$. Here a and b are called the **entry-** and **exit-time** of the boundary arc; a and b are also termed **junction points**. An arc is said to have a **contact point** with the boundary at $c \in (0, T)$ if $h(x(c), y(c), v(c)) = K$ and $h(x(t), y(t), v(t)) < K$ for all $t \neq c$ in a neighbourhood of c .

From (6) we have that on boundary arcs (where $y(t) = K$ for $t \in [a, b] \subset [0, T]$), the boundary control is determined by

$$\sigma_b(t) = \frac{1}{x(t)} \quad (7)$$

for $t \in (a, b)$ (see Maurer [17]). From Eq. (5b) $x'(t) = -\gamma K$ and therefore

$$\sigma_b(t) = \frac{1}{x(a) - \gamma K(t - a)} \text{ for } t \in (a, b). \quad (8)$$

In what follows we set forth Theorem 2 proved in [7] with $\kappa = 0$ for the unconstrained optimal control problem given by Eqs. (5a)–(5f). In that article, it is proven that an optimal control is bang-bang and has the form

$$\sigma[t, \mu](r) := \begin{cases} \sigma_f & \text{for } 0 \leq r \leq t, \\ \sigma_s & \text{for } t < r \leq t + \mu, \\ \sigma_f & \text{for } t + \mu < r \leq T, \end{cases} \quad (9)$$

with $t \geq 0$, $\mu \geq 0$ and $t + \mu \leq T$. In addition, it is defined $(x[t, \mu], y[t, \mu])$ as the associated state variables, solution of the system (5b)–(5c) with $\sigma = \sigma[t, \mu]$, and the functions

$$w(t) = \begin{cases} \int_t^{t+\tau} \frac{\sigma_f x[t, \tau](r) - 1}{y[t, \tau](r)} dr & \text{for } 0 \leq t \leq T - \tau \\ \int_t^T \frac{\sigma_f x[t, T-t](r) - 1}{y[t, T-t](r)} dr & \text{for } T - \tau < t \leq T, \end{cases} \quad (10)$$

and

$$\alpha(t) = \frac{1}{\gamma y[t, T-t](t)} \quad (11)$$

for $t \in [T - \tau, T]$.

Theorem 2.1 (Theorem 2 in [7] with $\kappa = 0$). Let $0 \leq \sigma_s < \sigma_f$ with $\sigma_s < 1$ and w be given by Eq. (10). Then the optimal control for the unconstrained problem given by Eqs. (5a)–(5f) is unique and is given by

$$\sigma^0(r) := \begin{cases} \sigma_f & \text{for } 0 \leq r \leq t^0, \\ \sigma_s & \text{for } t^0 < r \leq t^0 + \mu^0, \\ \sigma_f & \text{for } t^0 + \mu^0 < r \leq T, \end{cases} \quad (12)$$

where

1.1. For $w(0) \leq 0$: $t^0 = 0$ and $\mu^0 = \tau$.

1.2. For $w(0) > 0$ and $w(T - \tau) \leq 0$: $t^0 = \hat{t}$ and $\mu^0 = \tau$, where $\hat{t}$ is the unique value on $[0, T - \tau]$ such that $w(\hat{t}) = 0$.

1.3. For $0 < w(T - \tau) \leq \frac{1}{\gamma y[T - \tau, \tau](T - \tau)}$: $t^0 = T - \tau$ and $\mu^0 = \tau$.

1.4. For $w(T-\tau) > \frac{1}{\gamma y[T-\tau, \tau](T-\tau)}$: $t^0 = \tilde{t}$, where $\tilde{t}$ is the unique value on $[T-\tau, T]$ such that $w(\tilde{t}) = \frac{1}{\gamma y[\tilde{t}, T-\tilde{t}](\tilde{t})}$ and $\mu^0 = T - \tilde{t}$.

We now compute the partial derivatives of $x_\infty(x, y, \sigma)$ with respect to x and y in the same way that it is done in [5,7]. Recall that given $s > 0$ we previously defined $x_\infty(x, y, \sigma) = \lim_{t \rightarrow \infty} x(t)$, where $(x(t), y(t))$ is the solution of the system with initial condition (x, y) at time s and constant reproduction number $\sigma(t) \equiv \sigma$ for $t > s$.

Finally, recall that the solution of the SIR model without control (that is with constant $\sigma(t) = \sigma$), satisfies that

$$\rho(x(t), y(t), \sigma) := x(t)e^{-\sigma(x(t)+y(t))}$$

is constant in time and $y_\infty = 0$ (see for instance [15]). Then we have that $x_\infty = xe^{\sigma(x_\infty - x - y)} = \rho(x, y, \sigma)e^{\sigma x}$ and $w = -\sigma x_\infty$ satisfies the equation $w e^w = -\sigma \rho(x, y, \sigma)$. Therefore $w = W_0(-\sigma \rho(x, y, \sigma))$, where W_0 is the principal branch of Lambert's W -function [18], and thus for any $(x, y) \in D$

$$x_\infty(x, y, \sigma) = -\frac{1}{\sigma} W_0(\sigma \rho(x, y, \sigma)).$$

From this expression we can compute the partial derivatives of $x_\infty(x, y, \sigma)$ with respect to x and y (see [5]):

$$\frac{\partial x_\infty(x, y, \sigma)}{\partial x} = \frac{1 - \sigma x}{x} \frac{x_\infty(x, y, \sigma)}{1 - \sigma x_\infty(x, y, \sigma)}, \quad (13a)$$

$$\frac{\partial x_\infty(x, y, \sigma)}{\partial y} = -\frac{\sigma x_\infty(x, y, \sigma)}{1 - \sigma x_\infty(x, y, \sigma)}. \quad (13b)$$

3. Necessary conditions for an optimal control

To derive first order necessary optimality conditions, we apply Pontryagin's Maximum Principle for an optimal control problem with a pure state constraint. We assume that the state constraint is not active at the initial and final time, i.e., $y(0) < K, y(T) < K$. The corresponding augmented Hamiltonian H is given by

$$H(x, y, \sigma, \lambda) = -\lambda_1(\gamma \sigma x y) + \lambda_2(\gamma \sigma x y - \gamma y) + \lambda_3 \sigma \quad (14)$$

and the Lagrangian L

$$L(x, y, \sigma, \lambda, \eta) = H(x, y, \sigma, \lambda) + \eta(y - K), \quad (15)$$

where $\lambda \in \mathbb{R}^2$ and $\eta \geq 0$. We have omitted the state v since H is independent of this variable. Given an optimal solution $(x^*, y^*, v^*, \sigma^*)$, the necessary conditions for a maximum process $(x^*, y^*, v^*, \sigma^*)$ on $[0, T]$ are the following (see [16,19]):

There exists a real number $\lambda_0 \geq 0$, the adjoint variable $\lambda : [0, T] \rightarrow \mathbb{R}^3$, which is absolutely continuous, such that

$$(\lambda_0, \beta, \lambda(t), \eta(t)) \neq 0, \quad \text{for all } t \in [0, T] \quad (16)$$

and the following conditions hold:

1. The adjoint variables $\lambda_i(t)$ satisfy a.e. $t \in [0, T]$

$$\lambda_1'(t) = (\lambda_1(t) - \lambda_2(t))\gamma \sigma^*(t)y^*(t), \quad (17a)$$

$$\lambda_2'(t) = (\lambda_1(t) - \lambda_2(t))\gamma \sigma^*(t)x^*(t) + \gamma \lambda_2(t) + \eta(t) \quad (17b)$$

$$\lambda_3'(t) = 0, \quad (17c)$$

with final time conditions (using Eq. (13) and the abbreviation x_∞ for $x_\infty(x(T), y(T), \sigma_f)$)

$$\lambda_1(T) = \lambda_0 \frac{\partial x_\infty}{\partial x(T)} = \lambda_0 \frac{1 - \sigma_f x^*(T)}{x^*(T)} \frac{x_\infty}{1 - \sigma_f x_\infty}, \quad (17d)$$

$$\lambda_2(T) = \lambda_0 \frac{\partial x_\infty}{\partial y(T)} = -\lambda_0 \frac{\sigma_f x_\infty}{1 - \sigma_f x_\infty}. \quad (17e)$$

$$\lambda_3(T) = \beta \geq 0 \text{ and } \lambda_3(T)(v^*(T) - \sigma_f(T - \tau) - \sigma_s \tau) = 0. \quad (17f)$$

2. The function $\eta \geq 0$ satisfies

$$\eta(t)(y^*(t) - K) = 0, \quad (17g)$$

and is continuous on the interior of each boundary arc.

3. The jump condition at a contact point or junction point t is

$$\lambda_1(t^+) = \lambda_1(t^-), \quad (17h)$$

$$\lambda_2(t^+) = \lambda_2(t^-) + v, \quad v \geq 0, \quad (17i)$$

$$H(t^-) = H(t^+). \quad (17j)$$

4. For a.e. $t \in [0, T]$

$$H(x^*(t), y^*(t), \sigma^*(t), \lambda(t)) = \max_{\sigma: \sigma_s \leq \sigma \leq \sigma_f} H(x^*(t), y^*(t), \sigma, \lambda(t)).$$

We can rewrite

$$H(x^*, y^*, \sigma^*, \lambda) = \phi \sigma^* - \gamma \lambda_2 y$$

where

$$\phi(t) = \beta + \gamma x^*(t)y^*(t)(\lambda_2(t) - \lambda_1(t)), \quad (17k)$$

obtaining

$$\sigma^*(t)\phi(t) = \max_{\sigma: \sigma_s \leq \sigma \leq \sigma_f} \sigma \phi(t). \quad (17l)$$

5. There exists a constant C such that for a.e. $t \in [0, T]$

$$H(x^*(t), y^*(t), \sigma^*(t), \lambda(t)) = C,$$

that is

$$\phi(t)\sigma^*(t) - \gamma \lambda_2(t)y^*(t) = C. \quad (17m)$$

As the Hamiltonian is linear in $\sigma(t)$, the optimal control will depend on the sign of the switching function ϕ , given by (17k). If $\phi(t) = 0$ on a subinterval $I \subset [0, T]$, the control σ is called **singular** on I . Calling σ_{sing} the optimal control corresponding to a singular control and from the optimality condition (17l) we obtain the following lemma.

Lemma 3.1. Let σ^* be an optimal control, (x^*, y^*) the associated state given by (5b) and (5c) with $\sigma = \sigma^*$ and (λ_1, λ_2) solution of Eqs. (17a)–(17b) and (17d)–(17e). Then

$$\sigma^*(t) = \begin{cases} \sigma_f & \text{if } \phi(t) > 0 \\ \sigma_{sing}(t) & \text{if } \phi(t) = 0 \\ \sigma_s & \text{if } \phi(t) < 0. \end{cases} \quad (18)$$

We call (x^0, y^0) the associated optimal state given by the solution of system (5b)–(5c) with the optimal control $\sigma = \sigma^0$ as in Theorem 2.1. We consider two cases.

Case 1: $y^0(t) \leq K$ for all $t \in [0, T]$. This means that the optimal solution of problem (5a)–(5f) without the running state constraints, given by Theorem 2.1, satisfies also this latter condition (5g) implying that it is also an optimal solution of the complete problem (5). In this case we will have that there is no boundary arc.

Case 2: $\max_{[0, T]} y^0(t) > K$, that is, exists $t \in [0, T]$ such that $y^0(t) > K$.

The goal of the rest of this section is to prove Lemma 3.4 where we characterise the expression of an optimal control for the complete problem in Case 2, see Eq. (22). We begin by analysing the implications of the PMP conditions (17) on both interior and boundary arcs. This analysis provides insights into the adjoint variables λ_1 and λ_2 , which, in turn, will enable us to prove Eq. (23) from where we obtain (22) (see the proof of Lemma 3.4). Using (5b)–(5c) and (17a)–(17b) we first compute the derivative of ϕ for a.e. t :

$$\phi'(t) = \gamma x^*(t)y^*(t)(\gamma \lambda_1(t) + \eta(t)). \quad (19)$$

Interior arcs. Assume $(t_1, t_2) \subset [0, T]$ is an interior arc, that is $y^*(t) < K$ for all $t \in (t_1, t_2)$. Then, from (17g) we have that $\eta \equiv 0$ in (t_1, t_2) .

We now study the existence of singular controls for interior arcs. Assume $\phi(t) = 0$ on an interval $[a, b] \subset (t_1, t_2)$. Then using that $\eta \equiv 0$, from (19) we obtain

$$0 = \phi'(t) = \gamma^2 \lambda_1(t) x^*(t) y^*(t) \text{ a.e. } t \in (a, b), \quad (20)$$

from where we have that $\lambda_1 \equiv 0$ on (a, b) . Thus, from (17a) $\lambda_2 \equiv 0$ on (a, b) , concluding that $\lambda_1 \equiv \lambda_2 \equiv 0$ on (t_1, t_2) . As a consequence we have the following remark.

Remark 3.1. Given (t_1, t_2) an interior arc, if there exists $t \in (t_1, t_2)$ such that $(\lambda_1(t), \lambda_2(t)) \neq (0, 0)$, there cannot be singular arcs in (t_1, t_2) .

Boundary arcs. Assume $y^*(t) = K$ for all $t \in [t_1, t_2]$ and that the control corresponding to the boundary arc computed in (7) satisfies $\sigma_b(t) \in (\sigma_s, \sigma_f)$ for all $t \in (t_1, t_2)$. Since $\sigma_s < 1$ and $x(t) < 1$, from (7) the latter condition can be reduced to $\sigma_b(t) < \sigma_f$ for all $t \in (t_1, t_2)$. Then, from the optimality condition (17i) we deduce that $\phi(t) \equiv 0$ on (t_1, t_2) concluding that a boundary arc is a singular arc as shown in [17] for controls appearing linearly. From (19) we derive that $\eta(t) = -\gamma \lambda_1(t)$ on (t_1, t_2) and from (7) and (17b) we obtain $\lambda_2'(t) = 0$ on (t_1, t_2) .

Lemma 3.2. If (t_1, t_2) is a boundary arc, then $(t_1, t_2) \subset \{t \in [0, T] : \lambda_2(t) \leq \lambda_1(t) \leq 0\}$. Moreover we have that $(\lambda_1(t), \lambda_2(t)) \neq (0, 0)$ for all $t \in (t_1, t_2)$ or $\lambda_1 \equiv \lambda_2 \equiv 0$ on (t_1, t_2) .

Proof. Let (t_1, t_2) be a boundary arc. Since $\eta \geq 0$, from equality $\eta = -\gamma \lambda_1$ on boundary arcs we obtain $\lambda_1(t) \leq 0$ for all $t \in (t_1, t_2)$. In addition, for t such that $\lambda_1(t) < \lambda_2(t)$ it holds that $\phi(t) > 0$ implying that there cannot be singular arcs on $\{t \in [0, T] : \lambda_1(t) < \lambda_2(t)\}$ and thus neither boundary arcs. Moreover, from (17m) and using that $\phi(t) = 0$ for all $t \in (t_1, t_2)$, we have that $\lambda_2 \equiv -C/(\gamma K)$ is constant. Thus, if $(\lambda_1, \lambda_2)(s) = (0, 0)$ for some $s \in (t_1, t_2)$, we would have that $\lambda_2 \equiv 0$ on (t_1, t_2) and from Eq. (17a), λ_1 is a solution of the linear equation $\lambda_1'(t) = \gamma K \sigma_b(t) \lambda_1(t)$ on (t_1, t_2) with $\lambda_1(s) = 0$ concluding that $\lambda_1 \equiv 0$, yielding also $\eta \equiv 0$. $\square$

From the aforementioned we have that

$$\phi'(t) = \begin{cases} \gamma^2 \lambda_1(t) x^*(t) y^*(t) & \text{for nonsingular (interior) arcs} \\ 0 \text{ (and } \phi \equiv 0) & \text{for singular (interior or boundary) arcs} \end{cases} \quad (21)$$

In the next lemma we prove the normality of the optimal control problem (that is $\lambda_0 > 0$), the continuity of the adjoint variables and thus the continuity of the switching function ϕ and the existence of a boundary arc for **Case 2**.

Lemma 3.3. Assuming that for any contact or exit point s , $x^*(s) > \frac{1}{\sigma_f}$, then the switching function ϕ and λ_2 are continuous. In addition, if $y^*(T) < K$ the problem is normal ($\lambda_0 > 0$). Moreover, if $\max_{[0, T]} y^0(t) > K$, the constrained extremal has at least one boundary arc.

Proof. See Lemmas A.1–A.3 in the Supplement. $\square$

Using that $\lambda_0 > 0$, as it is usual, from now on we will consider $\lambda_0 = 1$. In the following lemma we prove the general expression of an optimal control for **Case 2**.

Lemma 3.4. If $\max_{[0, T]} y^0(t) > K$ and any contact or exit point s of the constraint extremal satisfies $x^*(s) > \frac{1}{\sigma_f}$, then the generalised optimal solution σ of problem (5) is given by

$$\sigma(t) = \begin{cases} \sigma_f & \text{for } 0 \leq t \leq t_1 \\ \sigma_b(t) = \frac{1}{x(t)} & \text{for } t_1 < t \leq t_2 \\ \sigma_s & \text{for } t_2 < t \leq t_3 \\ \sigma_f & \text{for } t_3 < t \leq T \end{cases} \quad (22)$$

where $0 \leq t_1 < t_2 \leq t_3 < T$, $t_1, t_2 \in \{t \in [0, T] : \lambda_2(t) \leq \lambda_1(t) \leq 0\}$ and $y(t) = K$ for all $t \in [t_1, t_2]$.

Proof. From Lemma 3.3 we have that the constraint extremal has at least one boundary arc and λ_2 and ϕ are continuous functions.

The proof follows by analysing the phase diagram of λ_1, λ_2 . We begin by noting that a solution (λ_1, λ_2) of the system (17a)–(17b) with final conditions (17d)–(17e) cannot cross both semilines $\lambda_1 = \lambda_2 \geq 0$ and $\lambda_1 = \lambda_2 < 0$ or $\lambda_1 = \lambda_2 > 0$ and $\lambda_1 = \lambda_2 \leq 0$. We prove the first case; the other is completely analogous. In fact, assume there exist $s_1, s_2 \in [0, T]$ such that $\lambda_1(s_1) = \lambda_2(s_1) \geq 0$ and $\lambda_1(s_2) = \lambda_2(s_2) < 0$. Evaluating (17k) on $t = s_i$ for $i = 1, 2$ we have that $\phi(s_i) = \beta \geq 0$ and from (17m), $\lambda_2(s_i) = \frac{\beta \sigma^*(s_i) - C}{\gamma y^*(s_i)}$. If $\beta = 0$ it holds that $\lambda_2(s_i) = -\frac{C}{\gamma y^*(s_i)}$ and if $\beta > 0$, using (18), we have that $\lambda_2(s_i) = \frac{\beta \sigma_f - C}{\gamma y^*(s_i)}$, both cases contradicting that $\lambda_2(s_1)$ and $\lambda_2(s_2)$ had opposite signs.

Since we have end time conditions on T we go backwards from $(\lambda_1(T), \lambda_2(T))$ with $\lambda_1(T) > 0$ and $\lambda_2(T) < 0$ (see (17d) and (17e)). Then, the solution (λ_1, λ_2) stays in the set $\{t \in [0, T] : \lambda_1(t) > 0, \lambda_2(t) < 0\}$ either for all $t \in [0, T]$ or for some maximal interval to the left of T , named $(T - \delta, T]$. For the latter case we have $\lambda_1(T - \delta) \leq 0$ or $\lambda_2(T - \delta) \geq 0$ and from Lemma 3.2, there are only interior arcs on $(T - \delta, T]$, concluding from (17g) that $\eta \equiv 0$ on $(T - \delta, T]$. Moreover, from Remark 3.1, we conclude that there cannot be singular arcs on $(T - \delta, T]$.

From (17a) and (17b), for $\lambda_2 = 0$ and $\lambda_1 > 0$, both λ_1 and λ_2 are increasing, showing that the trajectory backwards in time cannot go from the fourth to the first quadrant.

Since λ_1 is increasing for $\lambda_1 > \lambda_2$, the trajectory moves backwards in time to the left in the direction of the third quadrant. Moreover, it cannot touch the origin. In fact, assume $\lambda_1(T - \delta) = \lambda_2(T - \delta) = 0$, using that $\eta(t) = 0$ for $t \in (T - \delta, T]$ from (17a)–(17b), we would have that $\lambda_1(t) = \lambda_2(t) = 0$ for all $t \in [T - \delta, T]$ contradicting the end time conditions (17d)–(17e).

In case the trajectory crosses the semiline $\lambda_1 = 0, \lambda_2 < 0$, we call $\tilde{t} \in [0, T)$ such that $\lambda_1(\tilde{t}) = 0$. From (21) we would have that ϕ has a local minimum on $\tilde{t}$.

Also, on the region $\lambda_2 \leq \lambda_1 \leq 0$, from (21), ϕ is zero or a strictly decreasing function and from Lemma 3.3 there exists a boundary arc in this region.

For the aforementioned and (21) the generalised continuous function ϕ has the following structure (see Fig. 1)

$$\phi(t) = \begin{cases} \text{positive and decreasing} & \text{for } 0 \leq t \leq t_1 \\ 0 & \text{for } t_1 \leq t \leq t_2 \\ \text{negative and decreasing} & \text{for } t_2 \leq t \leq \tilde{t} \\ \text{negative and increasing} & \text{for } \tilde{t} \leq t \leq t_3 \\ \text{positive and increasing} & \text{for } t_3 \leq t \leq T, \end{cases} \quad (23)$$

where $0 \leq t_1 < t_2 \leq \tilde{t} \leq t_3 \leq T$ and $t_1, t_2 \in \{t \in [0, T] : \lambda_2(t) \leq \lambda_1(t) \leq 0\}$. Note that if $\phi(T) < 0$ (for instance when $\beta = 0$) then $t_3 = T$. The conclusion follows from Lemma 3.1.

4. Characterisation of the optimal control

In this section we will give the main result (Theorem 4.1) of this article. We recall that under the hypothesis of Lemma 3.4 and using the notation $t_3 = t_2 + \mu$ with $\mu \in [0, T - t_2]$, if σ is an optimal control, it has the form

$$\sigma[t_1, t_2, \mu](t) = \begin{cases} \sigma_f & \text{for } 0 \leq t \leq t_1 \\ \sigma_b(t) = \frac{1}{x(t)} & \text{for } t_1 < t \leq t_2 \\ \sigma_s & \text{for } t_2 < t \leq t_2 + \mu \\ \sigma_f & \text{for } t_2 + \mu < t \leq T \end{cases} \quad (24)$$

In order to determine the optimal solution we need to establish the times t_1^*, t_2^*, μ^* that make the control optimal.

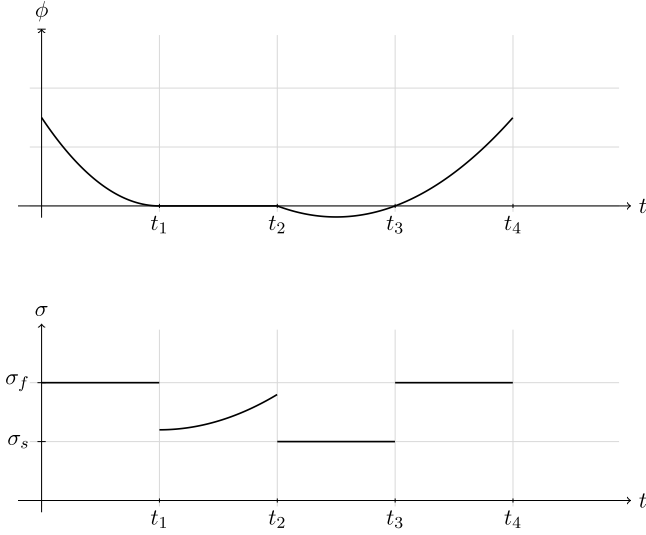


Fig. 1. Function ϕ and the corresponding σ given by (22).

For t_1, t_2, μ , let $x[t_1, t_2, \mu]$ be the associated state, solution of Eq. (5b) with $\sigma = \sigma[t_1, t_2, \mu]$. With the aim of analysing for which t_1, t_2, μ the integral restriction (5f) is satisfied, we compute the integral of $\sigma[t_1, t_2, \mu]$ on $[0, T]$ and clearing the variable μ we obtain a link between the parameters t_1, t_2 and μ given by

$$\mu \leq F[t_1](t_2), \quad (25)$$

where $F[t_1] : [t_1, T] \rightarrow \mathbb{R}$ is a function of t_2 defined by

$$F[t_1](t_2) := \tau + \frac{\sigma_f(t_2 - t_1)}{\sigma_s - \sigma_f} + \frac{1}{\gamma K(\sigma_s - \sigma_f)} \ln\left(1 - \frac{\gamma K(t_2 - t_1)}{x(t_1)}\right). \quad (26)$$

Note that since $x[t_1, t_2, \mu](t_2) = x(t_1) - \gamma K(t_2 - t_1)$ we have that the argument of the logarithm is positive showing that $F[t_1]$ is well defined.

Thus, the admissible controls $\sigma[t_1, t_2, \mu]$ given by (24) are characterised by the condition $(t_1, t_2, \mu) \in \mathcal{R}$ where

$$\mathcal{R} = \{(t_1, t_2, \mu) \in \mathbb{R}^3 : 0 \leq t_1 \leq t_2, 0 \leq \mu \leq F[t_1](t_2), \mu + t_2 \leq T\}. \quad (27)$$

In what follows, we will analyse **Case 2** (see the previous section); that is, we will assume there exists $t \in [0, T]$ such that $y^0(t) > K$. From Remark A.1 in the Supplement, y^0 attains its maximum at t^0 . Then, we can define

$$t_b = \min \{t \in [0, t^0] : y^0(t) = K\} \quad (28)$$

satisfying $t_b \in (0, t^0)$ where we have used that the initial condition $y_0 < K$.

In this case we will begin by proving that the time t_1^* that makes the control $\sigma[t_1^*, t_2, \mu]$ optimal for some t_2, μ is given by $t_1^* = t_b$ (see Lemma 4.1). Later we will study the region $\mathcal{R}$ (see Eq. (33)) and restrict the problem of finding (t_2^*, μ^*) for which $\sigma[t_1^*, t_2^*, \mu^*]$ is optimal.

Lemma 4.1. *If $\max_{[0, T]} y^0(t) > K$ and any contact or exit point s of the constraint extremal satisfies $x^*(s) > \frac{1}{\sigma_f}$, then the time $t_1^* = t_b$ with t_b defined in (28) makes the control $\sigma[t_1^*, t_2, \mu]$ optimal for some t_2, μ such that $(t_1^*, t_2, \mu) \in \mathcal{R}$.*

Proof. From (24) and the restriction (5g), it is clear that $t_1^* \leq t_b$. If the latter inequality is strict, there cannot be a boundary arc, contradicting Lemma A.2. $\square$

From now on, we will assume that t_1^* is fixed and given by $t_1^* = t_b$. It remains to characterise t_2^* and μ^* . In order to do that, we begin by restricting possible $t_2 > t_1^*$ for which $\sigma[t_1^*, t_2, \mu]$ is admissible.

Remark 4.1. In order for $\sigma[t_1^*, t_2, \mu]$ to be an admissible control, the boundary control must satisfy $\sigma_b(r) = 1/x[t_1^*, t_2, \mu](r) \in (\sigma_s, \sigma_f)$ for all $r \in (t_1^*, t_2)$. Then, since $x[t_1^*, t_2, \mu]$ is decreasing, the condition $x[t_1^*, t_2, \mu](t_2) \geq 1/\sigma_f$ has to be fulfilled and in particular $x[t_1^*, t_2, \mu](t) > 1/\sigma_f$ for all $t < t_2$. Using that for $t \in (t_1^*, t_2)$, $y[t_1^*, t_2, \mu](t) = K$ and $x[t_1^*, t_2, \mu](t) = -\gamma K(t - t_1^*) + x[t_1^*, t_2, \mu](t_1^*)$, we obtain that the condition $x[t_1^*, t_2, \mu](t_2) \geq 1/\sigma_f$ is satisfied if and only if $t_2 \leq t_1^* + \frac{x[t_1^*, t_2, \mu](t_1^*) - 1/\sigma_f}{\gamma K}$.

Then we compute, if there is any, the time t where the trajectory given by Eqs. (5b)–(5c) with $\sigma = \sigma[t_1^*, T, 0]$ satisfies that $x[t_1^*, T, 0](t) = 1/\sigma_f$ and define

$$t_m = \max \{t \in [t_1^*, T] : x[t_1^*, T, 0](t) \geq 1/\sigma_f\}. \quad (29)$$

Note that if $t_m < T$, then $t_m = t_1^* + \frac{x[t_1^*, T, 0](t_1^*) - 1/\sigma_f}{\gamma K}$. Also from (27), since $0 \leq \mu \leq F[t_1^*](t_2)$, we define, if there is any, the zero of $F[t_1^*]$ given by (26)

$$t_f = \max \{t \in [t_1^*, t_m] : F[t_1^*](t) \geq 0\} \quad (30)$$

and so we consider the restriction $F[t_1^*] : [t_1^*, t_f] \rightarrow \mathbb{R}$. Moreover, in what follows we analyse how conditions involving μ and t_2 on $\mathcal{R}$ in (27) are related. Using that $F[t_1^*](t_2) + t_2$ is increasing as a function of t_2 on $[t_1^*, t_f]$ (see Lemma A.5 in Supplement) we can define

$$t_c \in [t_1^*, t_f] \text{ such that } F[t_1^*](t_2) < T - t_2 \forall t_2 \in [t_1^*, t_c] \text{ and } F[t_1^*](t_2) > T - t_2 \forall t_2 \in (t_c, t_f]. \quad (31)$$

Note that if $F[t_1^*](t_2) > T - t_2$ for all $t_2 \in (t_1^*, t_f]$, then $t_c = t_1^*$ and if $F[t_1^*](t_2) < T - t_2$ for all $t_2 \in [t_1^*, t_f]$, then $t_c = t_f$.

From the definition of t_c we can define the continuous function $G : [t_1^*, t_f] \rightarrow \mathbb{R}$

$$G(t_2) = \min \{F[t_1^*](t_2), T - t_2\}, \quad (32)$$

which satisfies $G(t_c) = F[t_1^*](t_c) = T - t_c$ if $t_c \in (t_1^*, t_f)$. Note that if $t_c = t_1^*$, $G(t_2) = T - t_2$ and if $t_c = t_f$, $G(t_2) = F[t_1^*](t_2)$. Then for t_1^* fixed, the region $\mathcal{R}$ defined in (27) can be set as follows (see Fig. 2)

$$\begin{aligned} \mathcal{R}[t_1^*] &= \{(t_2, \mu) \in \mathbb{R}^2 : t_1^* \leq t_2 \leq t_f, 0 \leq \mu \leq F[t_1^*](t_2), \mu + t_2 \leq T\} \\ &= \{(t_2, \mu) \in \mathbb{R}^2 : t_2 \in [t_1^*, t_f], 0 \leq \mu \leq G(t_2)\}. \end{aligned} \quad (33)$$

Thus, we need to determine the maximum of the function

$$J(t_2, \mu) = x_\infty(x[t_1^*, t_2, \mu](T), y[t_1^*, t_2, \mu](T), \sigma_f) \quad (34)$$

with $(t_2, \mu) \in \mathcal{R}[t_1^*]$. In order to do that, we begin by computing the partial derivative of $J(t_2, \mu)$ with respect to μ and obtain (see Eq. (61) in the Supplement)

$$\frac{\partial J}{\partial \mu}(t_2, \mu) = \gamma y_3 \frac{x_\infty[t_1^*, t_2, \mu]}{1 - \sigma_f x_\infty[t_1^*, t_2, \mu]} (\sigma_f - \sigma_s) > 0, \quad (35)$$

where $x_\infty[t_1^*, t_2, \mu] = x_\infty(x[t_1^*, t_2, \mu](T), y[t_1^*, t_2, \mu](T), \sigma_f)$. From (35) we have that the optimal pair (t_2^*, μ^*) is attained at the superior border of the region $\mathcal{R}[t_1^*]$; that is, $\mu^* = G(t_2^*)$ with $t_2^* \in [t_1^*, t_f]$ and G defined in (32). Thus we consider J restricted to that border and define

$$\tilde{J}(t_2) := J(t_2, G(t_2)). \quad (36)$$

From (66) and (67) in the Supplement we obtain

$$\tilde{J}'(t_2) = \frac{\gamma^2 y_3 K(1 - \sigma_s x_2) x_\infty[t_1^*, t_2, G(t_2)]}{x_2(1 - \sigma_f x_\infty[t_1^*, t_2, G(t_2)])} \cdot \begin{cases} w_b(t_2) & \text{for } t_2 \in [t_1^*, t_c] \\ w_b(t_2) - \frac{1}{\gamma K} & \text{for } t_2 \in (t_c, t_f], \end{cases} \quad (37)$$

where $x_2 = x[t_1^*, t_2, G(t_2)](t_2)$, $y_3 = y[t_1^*, t_2, G(t_2)](t_2 + G(t_2))$ and $w_b : [t_1^*, t_f] \rightarrow \mathbb{R}$ is the continuous function

$$w_b(t_2) = \int_{t_2}^{t_2 + G(t_2)} \frac{\sigma_f x[t_1^*, t_2, G(t_2)](r) - 1}{y[t_1^*, t_2, G(t_2)](r)} dr. \quad (38)$$

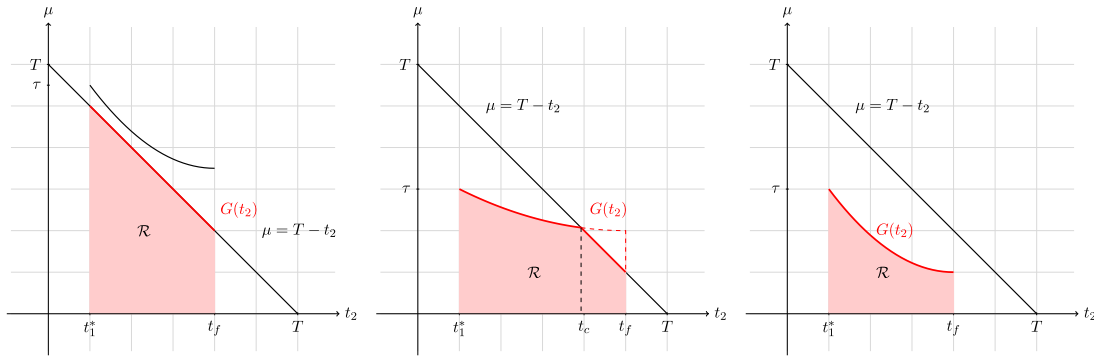


Fig. 2. Possible regions $\mathcal{R}$. For the region on the left, $F[t_1^*](t_2) > T - t_2 \forall t_2 \in [t_1^*, t_f]$, that is $t_c = t_1^*$. For the region in the centre, there exists $t_c \in (t_1^*, t_f)$ such that $F[t_1^*](t_c) = T - t_c$. Finally, for the region on the right, $F[t_1^*](t_2) < T - t_2 \forall t_2 \in [t_1^*, t_f]$, that is $t_c = t_f$.

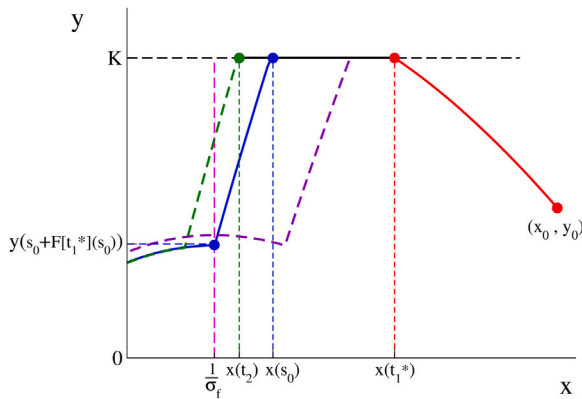


Fig. 3. Phase diagram of the trajectory $(x[t_1^*, t_2, F[t_1^*](t_2)], y[t_1^*, t_2, F[t_1^*](t_2)])(r)$ in red for $r \in [t_0, t_1^*]$ and in black for $r \in [t_1^*, t_2]$. The solid blue line for $r \in [s_0, s_0 + F[t_1^*](s_0)]$ corresponds to $t_2 = s_0$, while the dashed green and violet lines for $r \in [t_2, t_2 + F[t_1^*](t_2)]$ correspond to $t_2 > s_0$ and $t_2 < s_0$ respectively.

Next, we will study the monotonicity of the function $\tilde{J}(t_2)$ defined in (36). To do so, according to Eq. (37), we need to analyse the sign of the functions $w_b(t_2)$ for $t_2 \in [t_1^*, t_c]$ and $w_b(t_2) - \frac{1}{\gamma K}$. Remark 4.2 and Lemma 4.2 provide an answer for the behaviour of the first function, and Lemma 4.3 for the second. We will leave the proofs for the Supplement.

Remark 4.2. Since $x[t_1^*, t_2, F[t_1^*](t_2)](t_2 + F[t_1^*](t_2))$ is decreasing as a function of t_2 defined on the interval $[t_1^*, t_c]$ (see Lemma A.6 in Supplement), we can define $s_0 \in [t_1^*, t_c]$ such that $x[t_1^*, t_2, F[t_1^*](t_2)](t_2 + F[t_1^*](t_2)) > 1/\sigma_f$ for all $t_2 \in [t_1^*, s_0]$ and $x[t_1^*, t_2, F[t_1^*](t_2)](t_2 + F[t_1^*](t_2)) < 1/\sigma_f$ for all $t_2 \in (s_0, t_c]$. Note that if $x[t_1^*, t_2, F[t_1^*](t_2)](t_2 + F[t_1^*](t_2)) < 1/\sigma_f$ for all $t_2 \in (t_1^*, t_c]$, then $s_0 = t_1^*$ and if $x[t_1^*, t_2, F[t_1^*](t_2)](t_2 + F[t_1^*](t_2)) > 1/\sigma_f$ for all $t_2 \in [t_1^*, t_c]$, then $s_0 = t_c$. Then we can conclude that for $t_2 \in [t_1^*, s_0]$, $x[t_1^*, t_2, F[t_1^*](t_2)](t_2 + F[t_1^*](t_2)) > 1/\sigma_f$ and consequently $x[t_1^*, t_2, F[t_1^*](t_2)](r) \geq 1/\sigma_f$ for all $r \in (t_2, t_2 + F[t_1^*](t_2))$. See Fig. 3.

Lemma 4.2. For $s_0 \in [t_1^*, t_c]$ given by Remark 4.2, $w_b(t_2) > 0$ for all $t_2 \in [t_1^*, s_0]$ and w_b is a strictly decreasing function on (s_0, t_c) concluding that w_b changes sign at most once on $[t_1^*, t_c]$.

Lemma 4.3. On the interval $[t_c, t_f]$ the function $(w_b(t_2) - \frac{1}{\gamma K})y[t_1^*, t_2, T - t_2](T)$ is decreasing with respect to t_2 . Furthermore $w_b(t_f) \leq 0$ and $w_b(t_1^*) > 0$.

We are now in a position to state and prove the main result of this article.

Theorem 4.1. Let $0 \leq \sigma_s < \sigma_f$ with $\sigma_s < 1$. Let σ^0 be the optimal control for the unconstrained problem (5a)–(5f) given by (12) and (x^0, y^0) the associated optimal state given by the solution of the system (5b)–(5c) with $\sigma = \sigma^0$. In the case that there exists $t \in [0, T]$ such that $y^0(t) > K$ we assume that any contact or exit point s of the constraint extremal satisfies $x^*(s) > \frac{1}{\sigma_f}$. Then the optimal control σ^* for the restricted problem (5a)–(5g) is unique and is given by

$$\sigma^*(t) := \sigma[t_1^*, t_2^*, \mu^*](t) = \begin{cases} \sigma_f & \text{for } 0 \leq t \leq t_1^* \\ \sigma_b(t) & \text{for } t_1^* < t \leq t_2^* \\ \sigma_s & \text{for } t_2^* < t \leq t_2^* + \mu^* \\ \sigma_f & \text{for } t_2^* + \mu^* \leq t \leq T, \end{cases} \quad (39)$$

where $\sigma_b(t) = 1/x^*(t)$. Moreover,

Case 1: If $y^0(t) \leq K$ for all $t \in [0, T]$, then $t_1^* = t_2^* = t^0$, $\mu^* = \mu^0$ obtaining the result given in Theorem 2.1 where $\sigma[t^0, t^0, \mu^0] = \sigma^0$.

Case 2: If there exists $t \in [0, T]$ such that $y^0(t) > K$, then $t_1^* = t_b$ and for t_c defined in (31) and w_b defined in (38) we have

- 2.1. If $w_b(t_c) \leq 0$, then there exists a unique $\bar{t} \in (t_1^*, t_c)$ such that $w_b(\bar{t}) = 0$ and thus $t_2^* = \bar{t}$, $\mu^* = G(\bar{t})$.
- 2.2. If $0 < w_b(t_c) \leq \frac{1}{\gamma K}$, then $t_2^* = t_c$, $\mu^* = G(t_c)$.
- 2.3. If $w_b(t_c) > \frac{1}{\gamma K}$, then there exists a unique $\check{t} \in (t_c, t_f)$ such that $w_b(\check{t}) = \frac{1}{\gamma K}$ and thus $t_2^* = \check{t}$, $\mu^* = G(\check{t})$.

Proof. Let σ^0 and (x^0, y^0) be as in Theorem 2.1 and σ^* and (x^*, y^*, v^*) an optimal solution of the restricted problem. We consider the following cases:

Case 1: If $y^0(t) \leq K$ for all $t \in [0, T]$, it satisfies constraint (5g). Therefore σ^0 is an admissible control, concluding that is also optimal for the constrained problem. Thus, the optimal control consists of taking $t_2^* = t_1^* = t^0$ and $\mu^* = \mu^0$.

Case 2: If there exists $t \in [0, T]$ such that $y^0(t) > K$, from Lemma 4.1 we have that $t_1^* = t_b$. In what follows we will characterise $(t_2^*, \mu^*) \in \mathcal{R}[t_1^*]$ defined in (33).

- 2.1. Since $w_b(t_1^*) > 0$ (see Lemma 4.3) and $w_b(t_c) \leq 0$, from Lemma 4.2 there exists a unique $\bar{t} \in (t_1^*, t_c)$ such that $w_b(\bar{t}) = 0$, $w_b(s) > 0$ for $s \in [t_1^*, \bar{t})$ and $w_b(s) < 0$ for $s \in (\bar{t}, t_c]$. Thus, $\tilde{J}(t_2)$ increases on $[t_1^*, \bar{t})$ and, in the same way as for the previous item, from (37), we deduce that $\tilde{J}(t_2)$ decreases on $(\bar{t}, t_f]$ proving that $t_2^* = \bar{t}$ and $\mu^* = G(\bar{t})$.
- 2.2. If $0 < w_b(t_c) \leq \frac{1}{\gamma K}$, from Lemma 4.2 we have that $w_b(t_2) > 0$ for all $t_2 \in [t_1^*, t_c]$ and then, from (37), $\tilde{J}(t_2)$ is increasing on $[t_1^*, t_c]$. On the other hand, since $y[t_1^*, t_c, T - t_c](T) \left(w_b(t_c) - \frac{1}{\gamma K} \right) \leq 0$,

from Lemma 4.3 and Eq. (37), we have that $\tilde{J}(t_2)$ is decreasing on $[t_c, t_f]$ concluding that $t_2^* = t_c$ and $\mu^* = G(t_c)$.

- 2.3. If $w_b(t_c) > \frac{1}{\gamma K}$, from Lemma 4.2 we have that $w_b(t_2) > 0$ for all $t_2 \in [t_1^*, t_c]$. On the other hand, since $y[t_1^*, t_c, T - t_c](T) \left(w_b(t_c) - \frac{1}{\gamma K} \right) > 0$ from Lemma 4.3, we have that there exists a unique $\tilde{t} \in (t_c, t_f)$ such that $w_b(\tilde{t}) = \frac{1}{\gamma K}$. Thus, from (37), we have that $\tilde{J}(t_2)$ is increasing on $[t_1^*, \tilde{t}]$ and decreasing on $[\tilde{t}, t_f]$ proving that $t_2^* = \tilde{t}$ and $\mu^* = G(\tilde{t})$. $\square$

Remark 4.3. From the definition of t_c we have three possibilities: $t_c = t_1^*$, $t_c \in (t_1^*, t_f)$ or $t_c = t_f$. In the first case, when $t_c = t_1^*$, item 2.1. in Theorem 4.1 is never satisfied. Moreover, using a similar argument as in the proof of Lemma 4.3, from Lemma A.2 item 2.2. cannot be fulfilled obtaining $t_2^* = \tilde{t}$ and $\mu^* = G(\tilde{t}) = T - \tilde{t}$. If $t_c \in (t_1^*, t_f)$, the three items can be satisfied, for item 2.1. in Theorem 4.1 $G(t_2^*) = F[t_1^*](t_2^*)$ and for items 2.2. and 2.3. $G(t_2^*) = T - t_2^*$. Finally, in the third case, when $t_c = t_f$, from Lemma 4.3 we have that items 2.2. and 2.3. in Theorem 4.1 are never satisfied and thus we have $t_2^* = \tilde{t}$, $\mu^* = G(\tilde{t}) = F[t_1^*](\tilde{t})$.

5. Numerical results

Theorem 4.1 summarises the main results obtained in this article, which provides a rigorous optimal solution of the control problem defined in Eqs. (5). That is, Theorem 4.1 gives the optimal starting time of the strict quarantine t_2^* and its associated duration μ^* that minimises the cumulative number of ever-infected individuals (recovered), for a given set of epidemic parameters and under two restrictions: the maximum allowed fraction of infected individuals K over the control period T (5g), and (5f) implying that the maximum duration of the strict quarantine for the optimal solution is τ (see Lemma A.4). In particular, the optimal solutions can be grouped into two different case scenarios that depend on the relation between K and $y_{\max}^0$ (the maximum value of $y^0(t)$ in the interval $[0, T]$, where $y^0(t)$ is the optimum trajectory of the model without restriction). When $y_{\max}^0$ is larger than K , the optimum t_2^* is larger than the time t_1^* at which the trajectory of the system in the x - y space enters into the boundary arc $y(t) = K$, while for $y_{\max}^0$ smaller than K the system never enters into the boundary arc and the optimum t_2^* corresponds to that of the model with no restriction, $t_2^* = t^0$ and $\mu^* = \mu^0$ (Theorem 2.1).

In order to test Theorem 4.1, here we compare a numerical solution of the control problem (5) with that predicted by Theorem 4.1. We integrated the system of Eqs. (5b) and (5c) numerically under the constraints given by Eqs. (5f) and (5g), and found approximate values for the optima t_2^* and μ^* . Below we present results for two different thresholds, $K = 0.03$ (Figs. 4 and 5) and $K = 0.06$ (Figs. 6 and 7), which allow us to capture the different regimes corresponding to all possible different cases described in Theorem 4.1. For that, we calculated numerically $y_{\max}^0$ and the functions w and w_b , for several values of τ in the range $[0, 155]$, while keeping the other parameters fixed ($T = 365$, $\gamma = 0.1$, $\sigma_s = 0.8$, $\sigma_f = 1.5$), as shown in panels (c) and (d) of Figs. 4 and 6. Then, for a given τ , the values of $y_{\max}^0$, w and w_b determine the case as given in Theorem 4.1, and thus the numerical values of t_2^* and μ^* .

For the sake of concreteness, we chose the value $\gamma = 0.1/\text{day}$ for the recovery rate. This value of γ corresponds to a mean recovery time of 10 days, which falls within the range of Covid-19 estimates and sets the time scale of the system. All time scales are given in units of “days” in this section, even though we omit units for the sake of simplicity. Also, the strict quarantine and free quarantine reproduction numbers σ_s and σ_f , respectively, were chosen to fulfil the condition $\sigma_s < 1 < \sigma_f$, which tend to lead the system to disease extinction when $\sigma(t) = \sigma_s < 1$ and to disease outbreak when $\sigma(t) = \sigma_f > 1$.

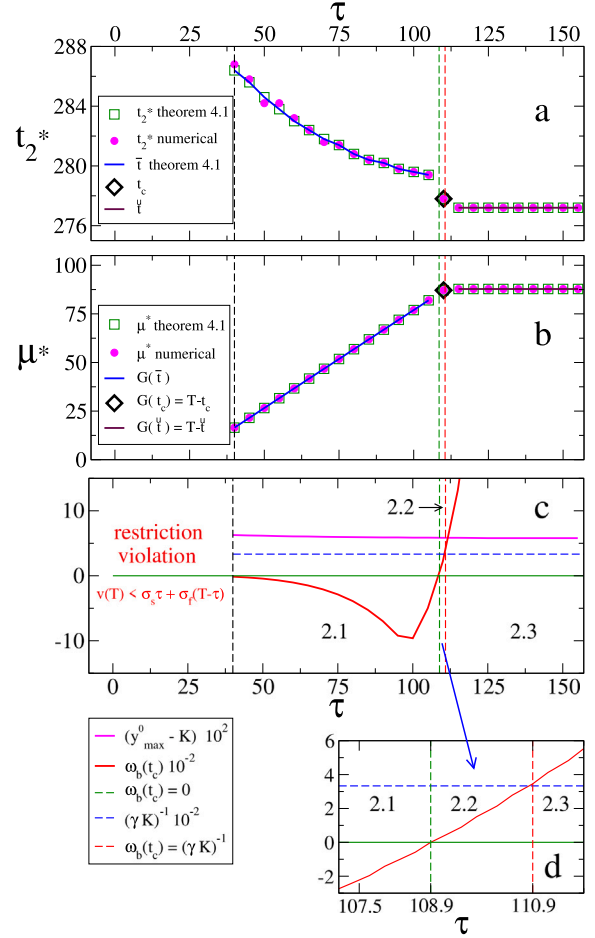


Fig. 4. Optimal solution for threshold $K = 0.03$, control period $T = 365$, recovery rate $\gamma = 0.1$, strict quarantine reproduction number $\sigma_s = 0.8$, and quarantine free reproduction number $\sigma_f = 1.5$. The initial condition is $x_0 = 1 - 10^{-6}$ and $y_0 = 10^{-6}$. (a) Optimal initial time of the strict quarantine t_2^* vs maximum allowed duration of the strict quarantine τ . (b) Optimal duration of the strict quarantine μ^* vs τ . In both panels (a) and (b), empty squares represent the exact solution given by Theorem 4.1, solid circles correspond to the numerical solution that maximises the asymptotic fraction of susceptible individuals x_∞ , and solid lines are the exact solution in each region. (c) Graphical determination of the times $\tau \approx 108.9$ and $\tau \approx 110.9$ denoted by vertical dashed lines that define the three regions for the different behaviours of t_2^* and μ^* . (d) Same as panel (c) with a smaller scale to see region 2.2 in more detail.

Fig. 4 shows cases 2.1, 2.2 and 2.3 of Theorem 4.1 that are observed for a relatively low threshold $K = 0.03$. For this value of K , $y_{\max}^0$ overcomes K for all τ values ($y_{\max}^0 > K$). These cases correspond to different intervals of τ determined by the functions $w_b(t_c)$ and $(\gamma K)^{-1}$, as we see in panels (c) and (d). In respective panels (a) and (b) we compare the behaviour of t_2^* and μ^* as a function of τ calculated numerically (filled circles) with that predicted from Theorem 4.1 (empty squares), where we observe an excellent agreement. The numerical estimation of the optimum (filled circles) was obtained by numerical integration of the dynamical system of equations ((5b) and (5c) with initial condition $(x_0, y_0) = (1 - 10^{-6}, 10^{-6})$ and constraints (5f) and (5g), for several sets of t_2 and μ , and calculating the values t_2^* and μ^* for which the asymptotic fraction of susceptible individuals x_∞ reaches a maximum value. For its part, the optimum from Theorem 4.1 (empty squares) was calculated according to the numerically estimated values of $\tilde{t}$, t_c and $\tilde{t}$ for t_2^* , and $G(\tilde{t}) = F[t_1^*](\tilde{t})$, $G(t_c) = T - t_c$ and $G(\tilde{t}) = T - \tilde{t}$ for μ^* , which correspond to cases 2.1, 2.2 and 2.3, respectively.

Right panels of Fig. 5 show the evolution of the system in the x - y space with threshold $K = 0.03$, for three different values of τ that correspond to the three different cases 2.1, 2.2 and 2.3, and various

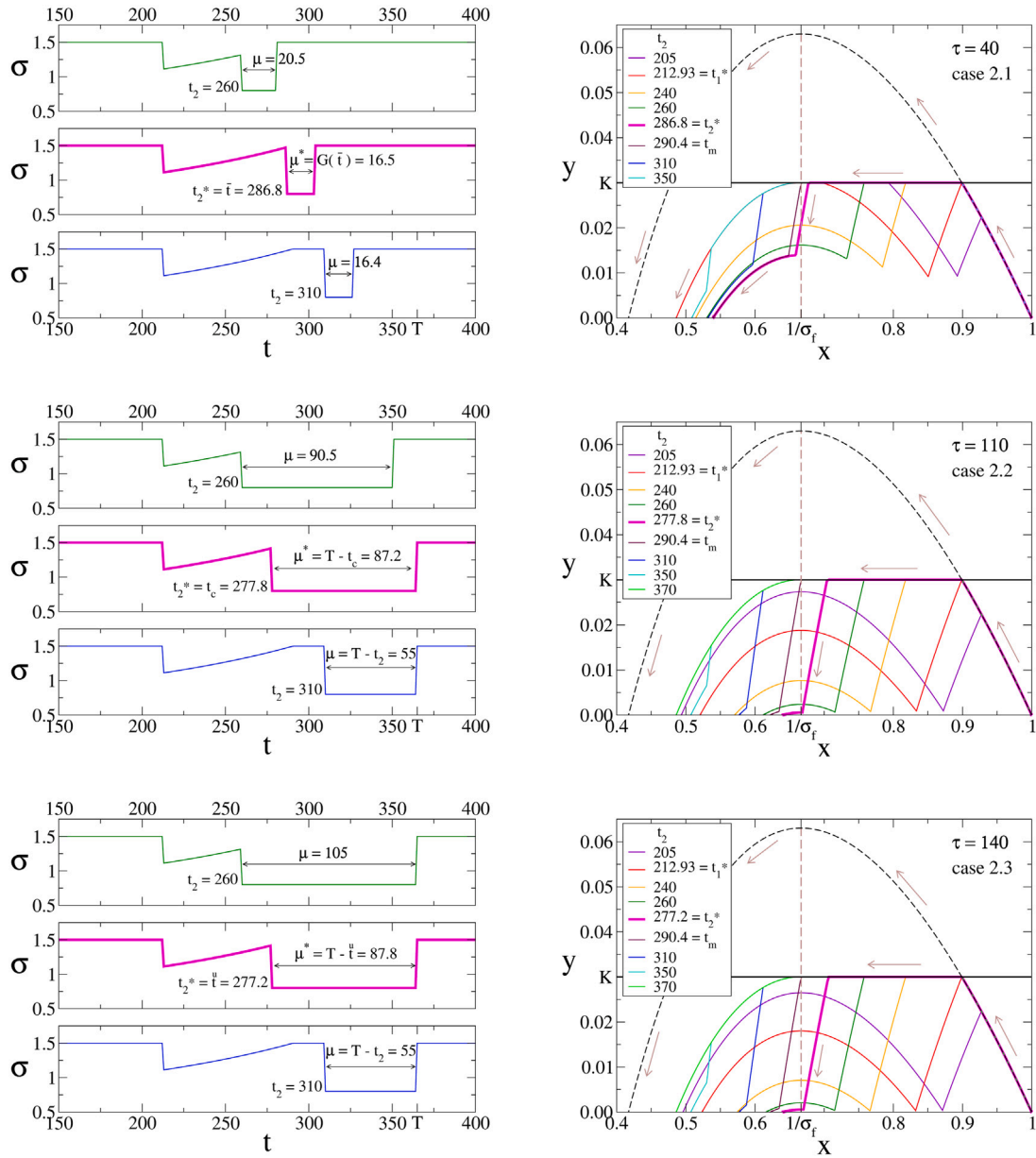


Fig. 5. System's trajectory in the x - y phase space (right panels) for $K = 0.03$, $T = 365$, $\gamma = 0.1$, $\sigma_s = 0.8$ and $\sigma_f = 1.5$, and the three values of τ indicated in the legends, corresponding to three different cases of the optimum time t_2^* , according to Theorem 4.1. Each individual trajectory corresponds to a different initial time of the strict quarantine t_2 . Pink lines denote the optimum trajectory, where the hitting time after the optimum time t_2^* after the hitting time $t_1^* \approx 212.93$ for a period μ^* (minimum of x_∞), while for the red and maroon trajectories the strict quarantine starts when the system hits ($t_2 = t_1^*$) and leaves ($t_2 = t_m \approx 290.4$) the boundary arc $y(t) = K$, respectively. The dashed black curves correspond to the no control case, where $y(t)$ clearly exceeds the threshold K . Left panels show the time evolution of $\sigma(t)$ for three different initial times t_2 in each case. The optimum times are $t_2^* \approx 286.8$ for $\tau = 40$ (top panels), $t_2^* \approx 277.8$ for $\tau = 110$ (middle panels), and $t_2^* \approx 277.2$ for $\tau = 140$ (bottom panels).

values of t_2 (beginning of strict quarantine $\sigma(t) = \sigma_s$) in each case. All curves $(x(t), y(t))$ start at (x_0, y_0) and initially follow the curve with quarantine free reproduction number $\sigma(t) = \sigma_f$. Then, some curves hit the boundary arc $y(t) = K$ at $t_1^* \approx 212.93$, and then continue along it with $\sigma(t) = 1/x(t)$ until they quit the boundary arc at either time t_2 when the strict quarantine begins, or t_m when $x(t_m) = 1/\sigma_f$. One curve shows the case when the strict quarantine is applied before the boundary arc ($t_2 = 205 < t_1^*$), and the rest of the cases correspond to applying the strict quarantine after t_m . In each curve the strict quarantine lasts a time μ that depends on t_2 and fulfils the restriction $v(T) \geq \sigma_s \tau + \sigma_f (T - \tau)$. The optimum (t_2^*, μ^*) leads to the maximum of x_∞ (pink curves). We see that in all cases the optimum corresponds to applying the strict quarantine after hitting the boundary arc ($t_2^* > t_1^*$) during a time period μ^* that is smaller than the maximum allowed τ . In

other words, it turns more efficient to start the strict quarantine while on the boundary arc, and keep it for a time shorter than the maximum allowed τ . We also show for comparison the trajectory corresponding to the uncontrolled case scenario $\sigma = \sigma_f$ (black dashed curves), where the proportion of infected individuals exceeds the threshold K associated to hospital capacity.

The left panels of Fig. 5 show the associated evolution of $\sigma(t)$ for the optimum control (t_2^*, μ^*) (pink line), and for other two values (t_2, μ) close to the optimum, for each case.

Top panels of Fig. 5 show the case 2.1 ($\tau = 40$), where the maximum of x_∞ is reached starting the strict quarantine at $t_2^* = \bar{t} \approx 286.8$ for a period $\mu^* = F[t_2^*](\bar{t}) \approx 16.5 < \tau$. Here we clearly see that implementing a shorter and later strict quarantine is more effective than using a longer but earlier strict quarantine, as we can see in the top-left panel

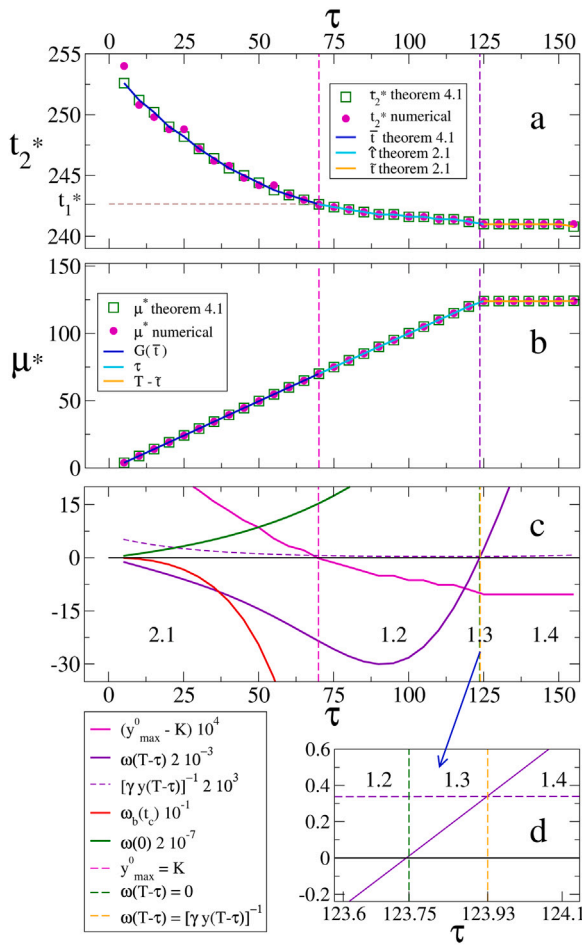


Fig. 6. Optimal solution for threshold $K = 0.06$, control period $T = 365$, recovery rate $\gamma = 0.1$, strict quarantine reproduction number $\sigma_s = 0.8$, and quarantine free reproduction number $\sigma_f = 1.5$. The initial condition is $x_0 = 1 - 10^{-6}$ and $y_0 = 10^{-6}$. (a) Optimal initial time of the strict quarantine t_2^* vs maximum allowed duration of the strict quarantine τ . (b) Optimal duration of the strict quarantine μ^* vs τ . In both panels (a) and (b), empty squares represent the exact solution (Theorem 4.1), solid circles correspond to the numerical solution that maximises x_∞ , and solid lines are the exact solution in each region. (c) Graphical determination of the times $\tau \approx 70$, $\tau \approx 123.75$ and $\tau \approx 123.93$ denoted by vertical dashed lines that define the four regions (cases) for the different behaviours of t_2^* and μ^* . (d) Same as panel (c) with a smaller scale to see region 1.3 in more detail.

by comparing $\sigma(t)$ for $\mu \approx 20.5$ (green curve) and $\mu^* \approx 16.5$ (pink curve). Middle panels show the case 2.2 ($\tau = 110$), where the optimum corresponds to $t_2^* = t_c \approx 277.8$ and $\mu^* = T - t_c \approx 87.2 < \tau$. Finally, bottom panels show the case 2.3 ($\tau = 140$), in which $t_2^* = \tilde{t} \approx 277.2$ and $\mu^* = T - \tilde{t} \approx 87.8 < \tau$.

The small threshold $K = 0.03$ used in the previous plots does not allow to see cases 1.2, 1.3 and 1.4 of Theorem 2.1. For that, we need to take a larger value of K , so that $y_{\max}^0 < K$ for τ large enough.

In Fig. 6 we show cases 2.1 and 1.2, 1.3 and 1.4 of Theorem 2.1 that are seen for a high threshold $K = 0.06$, where $y_{\max}^0 > K$ for $\tau \lesssim 70$ (case 2.1), while $y_{\max}^0 < K$ for $\tau \gtrsim 70$ (cases 1.2, 1.3 and 1.4). In Fig. 6(c) we observe that the region $\tau \lesssim 70$ corresponds to case 2.1 where $w_b(t_c) < 0$, and thus the optimum is given by $(t_2^*, \mu^*) = (\tilde{t}, G(\tilde{t}))$. Then, the region $70 \lesssim \tau \lesssim 123.75$ corresponds to case 1.2 ($w(T - \tau) < 0$) with an optimum at $(t_2^*, \mu^*) = (\hat{t}, \tau)$, the region $123.75 \lesssim \tau \lesssim 124.93$ is the case 1.3 ($0 < w(T - \tau) \leq (\gamma y(T - \tau))^{-1}$) with $(t_2^*, \mu^*) = (T - \tau, \tau)$, and the last region $\tau \gtrsim 123.93$ represents the case 1.4 ($w(T - \tau) > (\gamma y(T - \tau))^{-1}$), where $(t_2^*, \mu^*) = (\tilde{t}, T - \tilde{t})$. Fig. 7 is analogous to that of Fig. 5, where we show the trajectories (x, y) (right panels) and their associated evolution

of $\sigma(t)$ (left panels) for three values of τ corresponding to cases 1.2 ($\tau = 80$), 1.3 ($\tau = 123.8$) and 1.4 ($\tau = 150$), with optima (t_2^*, μ^*) at $(\hat{t}, \tau) \approx (242.2, 80)$, $(T - \tau, \tau) \approx (241.2, 123.8)$ and $(\tilde{t}, T - \tilde{t}) \approx (241.0, 124.0)$, respectively.

6. Summary and future work

In this work, we have analysed optimal quarantine measures using a time-dependent reproduction number within an SIR-type epidemic model. These control measures were considered as NPIs, ranging from partial lockdown to non-intervention.

The control strategy was derived based on necessary conditions provided by the Pontryagin Maximum Principle. This strategy is applied over a preset intervention period, aiming to minimise the final size of the epidemic while ensuring that medical care capacity and social and economic costs are not exceeded.

The optimal solution is a “wait, maintain, suspend, release” strategy (see Sereno [11] and Miclo [12]). It consists of at most four phases: an initial phase of no action with $\sigma^*(t) = \sigma_f$ that ends at time t_1^* , a second phase where the number of infected individuals is maintained at its maximum possible value $y^*(t) = K$ with $\sigma^*(t) = 1/x^*(t)$ that ends at t_2^* , followed by a third phase of partial lockdown with $\sigma^*(t) = \sigma_s$ applied over a single interval that ends at $t_2^* + \mu^*$, and concluding with a final phase of no intervention where $\sigma^*(t) = \sigma_f$ until the control time T . Subsequently, a specific optimal policy is deduced by optimising the transition times t_1^* , t_2^* , and $t_2^* + \mu^*$ of this generic strategy.

As demonstrated in Theorem 4.1 and tested with numerical simulations, the optimal time t_1^* to begin the maintain strategy depends on the initial values (x_0, y_0) and the threshold K . The optimal time t_2^* to finish the maintain strategy and begin the suspend phase also depends on the value τ , which restricts the duration of both the maintain and suspend strategies. With the addition of the peak prevalence constraint for all time, this result is an extension of the analytical result Theorem 2.1, proved in [7] for $\kappa = 0$ (meaning that the term in the objective functional involving the L^1 -norm for the control is absent).

Future work will focus on the consideration of more complex and realistic models. These models could combine different cost functionals from the literature (e.g., final size, L^1 -norm for the control, peak prevalence) with more general restrictions on the control and final state. In particular, including a term in the cost functional with the L^1 -norm for the control as studied in [7] would generalised the general result given in that article for $\kappa > 0$.

In addition, we could consider the bound K on the number of infected individuals to be an increasing time-dependent function, as might occur in an outbreak scenario (with the opening of new care centres).

CRedit authorship contribution statement

Rocío Balderrama: Writing – review & editing, Writing – original draft, Formal analysis. **Mariana Inés Prieto:** Writing – review & editing, Writing – original draft, Formal analysis. **Constanza Sánchez de la Vega:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Federico Vázquez:** Writing – review & editing, Validation, Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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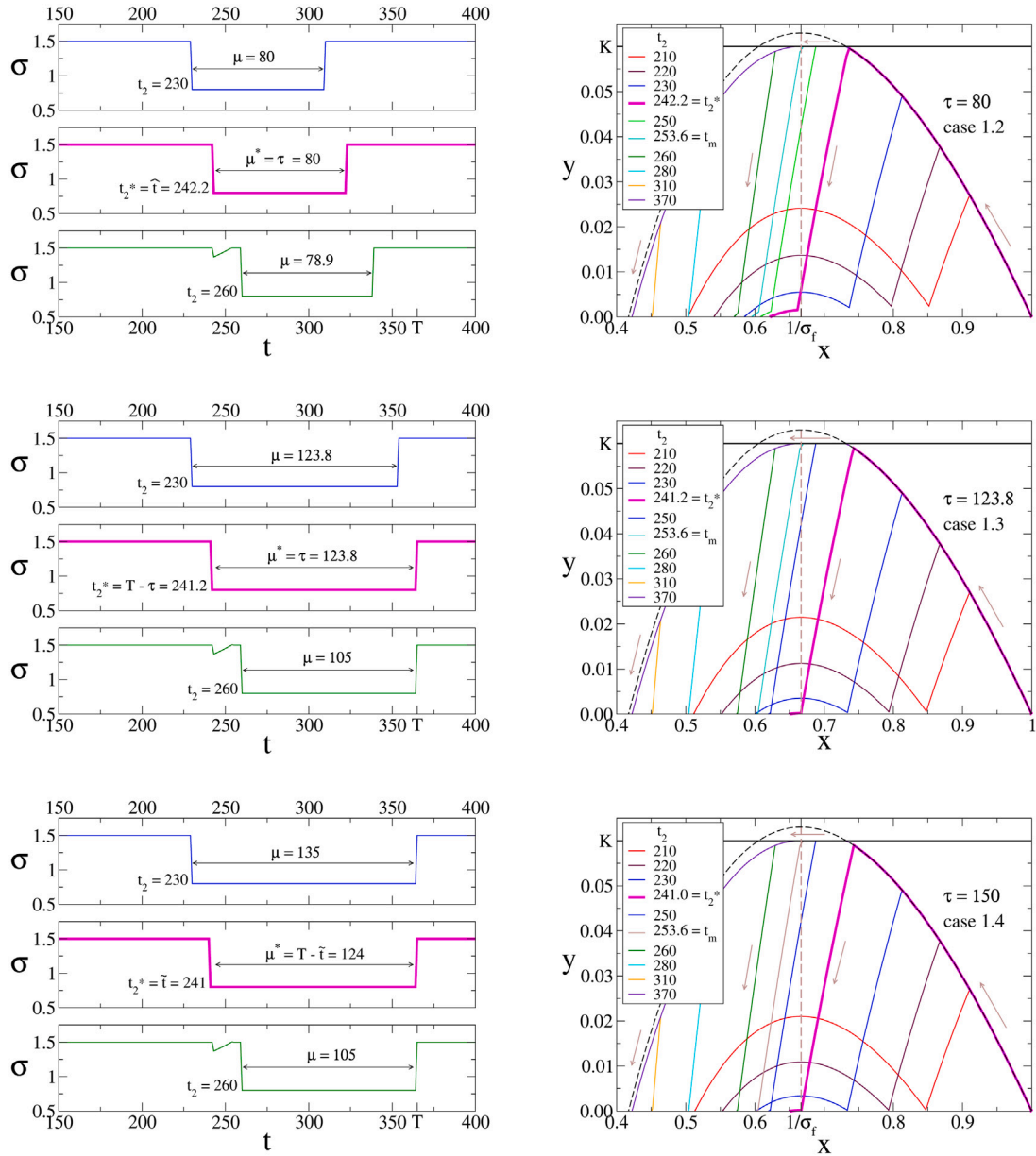


Fig. 7. System's trajectory in the x - y phase space (right panels) for $K = 0.06$, $T = 365$, $\gamma = 0.1$, $\sigma_s = 0.8$ and $\sigma_f = 1.5$, and the three values of τ indicated in the legends, corresponding to three different cases of the optimum time t_2^* (Theorem 4.1). Each individual trajectory corresponds to a different initial time of the strict quarantine t_2 . Pink lines denote the optimum trajectory, where the strict quarantine starts at the optimum time t_2^* before the hitting time $t_m^* \approx 242.63$ for a period μ^* (minimum of x_∞). For the maroon trajectory the strict quarantine starts when the system leaves the boundary arc $y(t) = K$ at $t_2 = t_m \approx 290.4$ ($x(t_m) = 1/\sigma_f$). The dashed black curves correspond to the no control case, where $y(t)$ clearly exceeds the threshold K . Left panels show the time evolution of $\sigma(t)$ for three different initial times t_2 in each case. The optimum times are $t_2^* \approx 242.2$ for $\tau = 80$ (top panels), $t_2^* \approx 241.2$ for $\tau = 123.8$ (middle panels), and $t_2^* \approx 241.0$ for $\tau = 150$ (bottom panels).

Appendix. Supplement

We begin with the proof of several results that make up the statement of Lemma 3.3 given in Section 3.

Remark A.1. Note that $y^0(t^0)$ is the global maximum of y^0 on $[0, T]$ and $t^0 \leq t_m$ where $x^0(t_m) = 1/\sigma_f$. In fact, from (5c) we have that $y(t)$ is decreasing for t such that $x(t) < 1/\sigma_f$, proving that $t^0 \leq t_m$. Moreover, for the cases 3. and 4. in Theorem 2.1, $t^0 + \mu^0 = T$ and therefore the optimal control $\sigma(t) = \sigma_f$ for $t \in [0, t^0]$ and $\sigma(t) = \sigma_s$ for $t \in (t^0, T]$. On the other hand, for the cases 1. and 2. observe that $w(t^0) \leq 0$, and thus, from (10) and (5b), it must hold $x^0(t^0 + \mu^0) \leq 1/\sigma_f$, from where we derive the desired result.

It is worth noticing that since $\sigma(t) \in [\sigma_s, \sigma_f]$ for all $t \in [0, T]$, from (5c) it holds $y'(t) < 0$ for all $t > r$ where $x(r) = 1/\sigma_f$, from where we derive that there cannot be any junction or contact point $s > r$.

Lemma A.1. Assuming that for any contact or exit point s , $x^*(s) > \frac{1}{\sigma_f}$, then the switching function ϕ and λ_2 are continuous.

Proof. From (17i) we have that for any junction or contact point s it holds:

$$\lambda_2(s^+) - \lambda_2(s^-) = v \quad (40)$$

with $v \geq 0$ implying

$$\phi(s^+) - \phi(s^-) = \gamma x^*(s) y^*(s) v \geq 0. \quad (41)$$

When s is a contact point such that $x^*(s) > \frac{1}{\sigma_f}$, from (18), it holds $\phi(s^-) \geq 0$ and $\phi(s^+) \leq 0$ respectively. Thus, from (41) we obtain $\phi(s^+) = \phi(s^-) = 0$. If s is an entry point, then $\phi(s^+) = 0$ implying from (41) that $\phi(s^-) \leq 0$. Assume $\phi(s^-) < 0$, then from (18), $\sigma(s^-) = \sigma_s$ and therefore $y'(s^-) < 0$ contradicting that s is an entry point. It follows that $\phi(s^+) = \phi(s^-) = 0$. Finally, if s is an exit point, then $\phi(s^-) = 0$ and from (41) $\phi(s^+) \geq 0$. Assume $\phi(s^+) > 0$, then from (18) $\sigma(s^+) = \sigma_f$ and therefore, using that $x^*(s) > \frac{1}{\sigma_f}$, we have $y'(s^+) > 0$ yielding a contradiction.

It follows that ϕ is continuous and therefore from (40) λ_2 is also a continuous function. $\square$

Moreover, we have the following lemma.

Lemma A.2. *If $\max_{[0,T]} y^0(t) > K$ and any contact point s of the constraint extremal satisfies $x^*(s) > \frac{1}{\sigma_f}$, then the constrained extremal has at least one boundary arc.*

Proof. If the constraint extremal satisfies $y^*(t) < K$ for all $t \in [0, T]$, then there are not any junction points and therefore the first order necessary conditions (17a)–(17m) hold for $\eta \equiv 0$ and λ_2 a continuous function. On the other hand, if the constraint extremal has only contact points, $\eta \equiv 0$ almost everywhere and from Lemma A.1, λ_2 is continuous. Thus, in both cases, the constraint extremal satisfies also the necessary conditions of the maximum principle for the unconstrained extremal, contradicting its uniqueness. This proves the desired result. $\square$

We now prove that the problem is normal.

Lemma A.3. *If the constraint extremal satisfies $y^*(T) < K$ and any contact point s satisfies $x^*(s) > \frac{1}{\sigma_f}$, then the optimal control problem (5) is normal, that is $\lambda_0 > 0$.*

Proof. Assume $\lambda_0 = 0$, then from (17d)–(17e), $\lambda_1(T) = \lambda_2(T) = 0$. Since $y^*(T) < K$, there exists $\delta > 0$ such that $y^*(t) < K$ for all $t \in (T - \delta, T]$ and therefore, from (17g), $\eta(t) = 0$ for all $t \in (T - \delta, T]$. Thus, from (17a)–(17b) we have $\lambda_1(t) = \lambda_2(t) = 0$ for all $t \in (T - \delta, T]$ and therefore, from (17k), $\phi(t) = \beta \geq 0$ for all $t \in (T - \delta, T]$. Using the non-triviality condition (16), we obtain $\beta > 0$ and from Lemma A.1, ϕ is a continuous function yielding there cannot be a boundary arc on any interval of the form $[r, T - \delta]$ since in this case we would have $\phi(t) = 0$ for all $t \in (r, T - \delta)$. Thus we conclude that $y^*(t) < K$ for almost every $t \in [0, T]$ implying $\phi(t) = \beta > 0$ for all $t \in [0, T]$. Then, from (18) we have $\sigma^*(t) = \sigma_f$ for all $t \in [0, T]$ contradicting the complementarity condition (17f). This proves the statement, then we can consider $\lambda_0 = 1$. $\square$

We now give some results concerning the analysis of the function $F[t_1^*](t_2)$ defined in (26) in Section 4.

Lemma A.4. *The function $F[t_1^*](t_2)$ defined in (26) is strictly decreasing. Moreover for $(\mu, t_2) \in \mathcal{R}[t_1^*]$ defined in (33) it holds*

$$\mu \leq F[t_1^*](t_2) \leq Ft_1^* = \tau \text{ for all } t_2 \in [t_1^*, t_f]. \quad (42)$$

Proof. We compute the derivative

$$\frac{dF[t_1^*]}{dt_2}(t_2) = \frac{1}{\sigma_s - \sigma_f} \left(\sigma_f - \frac{1}{x[t_1^*, t_2, \mu](t_2)} \right). \quad (43)$$

Using that $\sigma_s - \sigma_f < 0$ and $\sigma_f - \frac{1}{x[t_1^*, t_2, \mu](t_2)} > 0$ (see Remark 4.1), we obtain that F is strictly decreasing. Therefore we obtain (42). $\square$

Lemma A.5. *Let $H : [t_1^*, t_f] \rightarrow \mathbb{R}$ be the function defined by $H(t_2) = F[t_1^*](t_2) + t_2$. Then H is an increasing function.*

Proof. Computing the derivative of $H(t_2)$ we obtain

$$H'(t_2) = 1 + \frac{dF[t_1^*]}{dt_2}(t_2) = \frac{1}{\sigma_s - \sigma_f} \left(\sigma_s - \frac{1}{x[t_1^*, t_2, \mu](t_2)} \right) > 0. \quad \square \quad (44)$$

Next, we will do some preliminary calculations in order to compute the partial derivative $\frac{\partial J}{\partial \mu}(t_2, \mu)$ given by (35) and also the derivative of $\tilde{J}(t_2)$ given by (37) and its monotonicity.

We denote the solution of system

$$x'(r) = -\gamma \sigma(r)x(r)y(r), \quad x(t) = x \quad (45a)$$

$$y'(r) = \gamma \sigma(r)x(r)y(r) - \gamma y(r), \quad y(t) = y \quad (45b)$$

for $r \geq t$ by

$$\Psi(r, t, x, y, \sigma), \text{ with } \sigma \equiv \sigma_s, \sigma \equiv \sigma_f \text{ or } \sigma(r) = \sigma_b(r) = 1/x(r)$$

$$\text{and initial data } (x, y) \in \mathcal{D} \text{ at time } t. \quad (46)$$

If we denote $(x_1, y_1) = \Psi(t_1^*, t_0, x_0, y_0, \sigma_f)$, $(x_2, y_2) = \Psi(t_2, t_1^*, x_1, y_1, \sigma_b)$, $(x_3, y_3) = \Psi(t_2 + \mu, t_2, x_2, y_2, \sigma_s)$, then, from (24)

$$(x[t_1^*, t_2, \mu](r), y[t_1^*, t_2, \mu](r)) = \begin{cases} \Psi(r, t_0, x_0, y_0, \sigma_f) & \text{for } 0 \leq r \leq t_1^* \\ \Psi(r, t_1^*, x_1, y_1, \sigma_b) & \text{for } t_1^* < r \leq t_2 \\ \Psi(r, t_2, x_2, y_2, \sigma_s) & \text{for } t_2 < r \leq t_2 + \mu \\ \Psi(r, t_2 + \mu, x_3, y_3, \sigma_f) & \text{for } t_2 + \mu < r \leq T \end{cases} \quad (47)$$

While the majority of the subsequent content is covered in the Supplement of [7], we will incorporate certain proofs essential to ensure the self-sufficiency of this article.

We recall two properties for the solutions of ordinary differential equations. First, the relation between the derivative with respect to initial time and the derivatives with respect to initial data give us the equation

$$\frac{\partial \Psi_j(s, t, x, y, \sigma)}{\partial t} = \frac{\partial \Psi_j(s, t, x, y, \sigma)}{\partial x} \gamma \sigma x y - \frac{\partial \Psi_j(s, t, x, y, \sigma)}{\partial y} \gamma y (\sigma x - 1) \quad (48)$$

for Ψ defined in (46), with $s \geq t$, $\sigma \in \{\sigma_s, \sigma_f\}$, initial data $(x, y) \in \mathcal{D}$ at time t and $j = 1, 2$.

Second, the dependence of the solution $\Psi(s, t, x, y, \sigma)$ with respect to initial data x, y is given by the following known equations. For simplicity of notation, when there is no risk of confusion, we will denote $\Psi(s)$ for $\Psi(s, t, x, y, \sigma)$,

$$\begin{pmatrix} \frac{\partial \Psi_1}{\partial x} & \frac{\partial \Psi_1}{\partial y} \\ \frac{\partial \Psi_2}{\partial x} & \frac{\partial \Psi_2}{\partial y} \end{pmatrix}'(s) = \begin{pmatrix} -\gamma \sigma \Psi_2(s) & -\gamma \sigma \Psi_1(s) \\ \gamma \sigma \Psi_2(s) & \gamma (\sigma \Psi_1(s) - 1) \end{pmatrix} \cdot \begin{pmatrix} \frac{\partial \Psi_1}{\partial x} & \frac{\partial \Psi_1}{\partial y} \\ \frac{\partial \Psi_2}{\partial x} & \frac{\partial \Psi_2}{\partial y} \end{pmatrix}(s). \quad (49)$$

with initial data

$$\begin{pmatrix} \frac{\partial \Psi_1}{\partial x} & \frac{\partial \Psi_1}{\partial y} \\ \frac{\partial \Psi_2}{\partial x} & \frac{\partial \Psi_2}{\partial y} \end{pmatrix}(t) = I d. \quad (50)$$

Then, denoting $t_3 = t_2 + \mu$, $t_4 = T$, $\sigma_2 = \sigma_s$ and $\sigma_3 = \sigma_f$, for $i = 2, 3$, we define

$$u_i(s) = u(s, t_i, x_i, y_i, \sigma_i) = \frac{\partial \Psi_1}{\partial x_i}(s, t_i, x_i, y_i, \sigma_i) - \frac{\partial \Psi_1}{\partial y_i}(s, t_i, x_i, y_i, \sigma_i), \quad (51a)$$

$$v_i(s) = v(s, t_i, x_i, y_i, \sigma_i) = \frac{\partial \Psi_2}{\partial x_i}(s, t_i, x_i, y_i, \sigma_i) - \frac{\partial \Psi_2}{\partial y_i}(s, t_i, x_i, y_i, \sigma_i), \quad (51b)$$

for $s \in [t_i, t_{i+1}]$, that satisfy the system of equations on u_i and v_i

$$\begin{pmatrix} u_i'(s) \\ v_i'(s) \end{pmatrix} = \begin{pmatrix} -\gamma \sigma_i \Psi_2(s) & -\gamma \sigma_i \Psi_1(s) \\ \gamma \sigma_i \Psi_2(s) & \gamma (\sigma_i \Psi_1(s) - 1) \end{pmatrix} \cdot \begin{pmatrix} u_i(s) \\ v_i(s) \end{pmatrix}. \quad (52)$$

with initial data

$$\begin{pmatrix} u_i(t_i) \\ v_i(t_i) \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad (53)$$

Moreover, using that for any $(x_i, y_i) \in \mathcal{D}$, $\Psi(s, t_i, x_i, y_i, \sigma_i)$, satisfies for $s \in [t_i, t_{i+1}]$

$$\Psi_1(s, t_i, x_i, y_i, \sigma_i) e^{-\sigma_i(\Psi_1(s, t_i, x_i, y_i, \sigma_i) + \Psi_2(s, t_i, x_i, y_i, \sigma_i))} = x_i e^{-\sigma_i(x_i + y_i)},$$

we compute the derivatives with respect to x_i and y_i for $s \in [t_i, t_{i+1}]$

$$\frac{\partial \Psi_1}{\partial x_i}(s) - \sigma_i \Psi_1(s) \left(\frac{\partial \Psi_1}{\partial x_i}(s) + \frac{\partial \Psi_2}{\partial x_i}(s) \right) = (1 - \sigma_i x_i) \frac{\Psi_1(s)}{x_i}, \quad (54a)$$

$$\frac{\partial \Psi_1}{\partial y_i}(s) - \sigma_i \Psi_1(s) \left(\frac{\partial \Psi_1}{\partial y_i}(s) + \frac{\partial \Psi_2}{\partial y_i}(s) \right) = -\sigma_i \Psi_1(s). \quad (54b)$$

where, for simplicity of notation, we have called $\Psi_j(s) = \Psi_j(s, t_i, x_i, y_i, \sigma_i)$ for $j = 1, 2$ and $s \in [t_i, t_{i+1}]$. Then, subtracting the last two equations

$$u_i(s) - \sigma_i \Psi_1(s, t_i, x_i, y_i, \sigma_i)(u_i(s) + v_i(s)) = \frac{\Psi_1(s, t_i, x_i, y_i, \sigma_i)}{x_i} \quad (55)$$

for $s \in [t_i, t_{i+1}]$ and therefore using (55), from (52) we have that u_i satisfies the ordinary differential equation

$$u_i'(s) = \gamma \left(\sigma_i \Psi_1(s, t_i, x_i, y_i, \sigma_i) - \sigma_i \Psi_2(s, t_i, x_i, y_i, \sigma_i) - 1 \right) u_i(s) + \gamma \frac{\Psi_1(s, t_i, x_i, y_i, \sigma_i)}{x_i},$$

$$u_i(t_i) = 1.$$

Thus, for $s \in [t_i, t_{i+1}]$ when $i = 2, 3$, we obtain

$$\begin{aligned} u_i(s) &= \frac{\Psi_1(s) \Psi_2(s)}{x_i y_i} + \gamma \frac{\Psi_1(s) \Psi_2(s)}{x_i} \int_{t_i}^s \frac{1}{\Psi_2(r)} dr \\ &= \frac{\Psi_1(s)}{x_i} + \gamma \sigma_i \frac{\Psi_1(s) \Psi_2(s)}{x_i} \int_{t_i}^s \frac{\Psi_1(r)}{\Psi_2(r)} dr, \end{aligned} \quad (56)$$

$$v_i(s) = -\frac{\Psi_1(s)}{x_i} + \frac{(1 - \sigma_i \Psi_1(s)) \Psi_2(s)}{x_i} \gamma \int_{t_i}^s \frac{\Psi_1(r)}{\Psi_2(r)} dr, \quad (57)$$

$$u_i(s) + v_i(s) = \frac{\Psi_2(s)}{x_i} \gamma \int_{t_i}^s \frac{\Psi_1(r)}{\Psi_2(r)} dr. \quad (58)$$

From (47), using (48) and (51), after some computations we have, for $r \in (t_2 + \mu, T]$

$$\frac{\partial x}{\partial \mu}[t_1^*, t_2, \mu](r) = \gamma x_3 y_3 (\sigma_f - \sigma_s) u_3(r) \quad (59a)$$

$$\frac{\partial y}{\partial \mu}[t_1^*, t_2, \mu](r) = \gamma x_3 y_3 (\sigma_f - \sigma_s) v_3(r) \quad (59b)$$

and for $r \in [0, t_2 + \mu]$

$$\frac{\partial x}{\partial \mu}[t_1^*, t_2, \mu](r) = 0, \quad \frac{\partial y}{\partial \mu}[t_1^*, t_2, \mu](r) = 0. \quad (60)$$

Therefore, combining (13) with (59) for $r = T$ and (55) for $i = 3$ we obtain

$$\frac{\partial J}{\partial \mu}(t_2, \mu) = \gamma y_3 (\sigma_f - \sigma_s) \frac{x_\infty[t_1^*, t_2, \mu]}{1 - \sigma x_\infty[t_1^*, t_2, \mu]} \quad (61)$$

where $x_\infty[t_1^*, t_2, \mu] = x_\infty(x[t_1^*, t_2, \mu](T), y[t_1^*, t_2, \mu](T), \sigma_f)$. In what follows we will compute $\bar{J}'(t_2)$. We begin by computing the derivative of $x[t_1^*, t_2, \mu](r)$ with respect to t_2 . From (47), we have that for $r \in [0, t_2]$,

$$\frac{\partial x}{\partial t_2}[t_1^*, t_2, \mu](r) = 0, \quad \frac{\partial x}{\partial t_2}[t_1^*, t_2, \mu](r) = 0.$$

For $r \in (t_2, t_2 + \mu)$, using (48), (51) and the fact that $x_2 = x[t_1^*, t_2, \mu](t_2^*) - \gamma K(t_2 - t_1)$ and $y_2 = K$, we obtain

$$\frac{\partial x}{\partial t_2}[t_1^*, t_2, \mu](r) = \gamma K (\sigma_s x_2 - 1) u_2(r) \quad (62a)$$

$$\frac{\partial y}{\partial t_2}[t_1^*, t_2, \mu](r) = \gamma K (\sigma_s x_2 - 1) v_2(r) \quad (62b)$$

Finally, for $r \in (t_2 + \mu, T]$, from (47), using (45) and (46) with $\sigma = \sigma_s$ and initial data (x_2, y_2) at time t_2 , (48) and (51) we obtain

$$\frac{\partial x_3}{\partial t_2} = -\gamma \sigma_s x_3 y_3 + \gamma K (\sigma_s x_2 - 1) u_2(t_2 + \mu) \quad (63a)$$

$$\frac{\partial y_3}{\partial t_2} = \gamma y_3 (\sigma_s x_3 - 1) + \gamma K (\sigma_s x_2 - 1) v_2(t_2 + \mu). \quad (63b)$$

Thus,

$$\frac{\partial x}{\partial t_2}[t_1^*, t_2, \mu](r) = \gamma x_3 y_3 (\sigma_f - \sigma_s) u_3(r) + \gamma K (\sigma_s x_2 - 1) \cdot$$

$$\left[\frac{\partial \Psi_1}{\partial x_3}(r, t_2 + \mu, x_3, y_3, \sigma_f) u_2(t_2 + \mu) + \frac{\partial \Psi_1}{\partial y_3}(r, t_2 + \mu, x_3, y_3, \sigma_f) v_2(t_2 + \mu) \right]$$

and

$$\frac{\partial y}{\partial t_2}[t_1^*, t_2, \mu](r) = \gamma x_3 y_3 (\sigma_f - \sigma_s) v_3(r) + \gamma K (\sigma_s x_2 - 1)$$

$$\left[\frac{\partial \Psi_2}{\partial x_3}(r, t_2 + \mu, x_3, y_3, \sigma_f) u_2(t_2 + \mu) + \frac{\partial \Psi_2}{\partial y_3}(r, t_2 + \mu, x_3, y_3, \sigma_f) v_2(t_2 + \mu) \right].$$

After some computations, combining (13) with (54) and (55) for $s = T$ and for $s = t_3$ we obtain

$$\begin{aligned} \frac{\partial J}{\partial t_2}(t_2, \mu) &= \frac{x_\infty[t_1^*, t_2, \mu]}{(1 - \sigma_f x_\infty[t_1^*, t_2, \mu])} \\ &\quad \left[\gamma (\sigma_f - \sigma_s) y_3 + \gamma K (\sigma_s x_2 - 1) \right. \\ &\quad \times \left. \left(\frac{1}{x_2} - (\sigma_f - \sigma_s)(u_2(t_2 + \mu) + v_2(t_2 + \mu)) \right) \right]. \end{aligned}$$

From (58) and using that integrating $\left(\frac{1}{y[t_1^*, t_2, \mu]} \right)'(r)$ on the interval $[t_2, t_3]$ (see equations (74) and (75) in [7] for $i = 2$ and $s = t_3$) we obtain

$$\begin{aligned} 1 + \gamma y_3 (\sigma_s - \sigma_f) \int_{t_2}^{t_2 + \mu} \frac{x[t_1^*, t_2, \mu](r)}{y[t_1^*, t_2, \mu](r)} dr \\ = \frac{y_3}{K} - \gamma y_3 \int_{t_2}^{t_2 + \mu} \frac{\sigma_f x[t_1^*, t_2, \mu](r) - 1}{y[t_1^*, t_2, \mu](r)} dr, \end{aligned} \quad (64)$$

we conclude that

$$\begin{aligned} \frac{\partial J}{\partial t_2}(t_2, \mu) &= \frac{\gamma y_3 x_\infty[t_1^*, t_2, \mu]}{(1 - \sigma_f x_\infty[t_1^*, t_2, \mu])} \\ &\quad \times \left[\sigma_f - \frac{1}{x_2} - \gamma K \left(\sigma_s - \frac{1}{x_2} \right) \int_{t_2}^{t_2 + \mu} \frac{\sigma_f x[t_1^*, t_2, \mu](r) - 1}{y[t_1^*, t_2, \mu](r)} dr \right]. \end{aligned} \quad (65)$$

Therefore, from (65), (61) and (43) we obtain for $t_2 \in [t_1^*, t_c]$

$$\begin{aligned} \frac{\partial}{\partial t_2} J(t_2, F[t_1^*](t_2)) &= \frac{\gamma^2 y_3 K (1 - \sigma_s x_2) x_\infty[t_1^*, t_2, F[t_1^*](t_2)]}{x_2 (1 - \sigma_f x_\infty[t_1^*, t_2, F[t_1^*](t_2)])} \\ &\quad \times \int_{t_2}^{t_2 + F[t_1^*](t_2)} \frac{\sigma_f x[t_1^*, t_2, F[t_1^*](t_2)](r) - 1}{y[t_1^*, t_2, F[t_1^*](t_2)](r)} dr, \end{aligned} \quad (66)$$

and for $t_2 \in (t_c, t_f]$

$$\begin{aligned} \frac{\partial}{\partial t_2} J(t_2, T - t_2) &= \frac{\gamma^2 y_3 K (1 - \sigma_s x_2) x_\infty[t_1^*, t_2, T - t_2]}{x_2 (1 - \sigma_f x_\infty[t_1^*, t_2, T - t_2])} \\ &\quad \times \left(\int_{t_2}^T \frac{\sigma_f x[t_1^*, t_2, T - t_2](r) - 1}{y[t_1^*, t_2, T - t_2](r)} dr - \frac{1}{\gamma K} \right). \end{aligned} \quad (67)$$

Lemma A.6. The function $\varphi : [t_1^*, t_c] \rightarrow \mathbb{R}$

$$\varphi(t_2) = \Psi_1(t_2 + F[t_1^*](t_2), t_2, x_2, y_2, \sigma_s)$$

is decreasing.

Proof. Using similar computations to (63a) combined with (44) we obtain

$$\varphi'(t_2) = (\sigma_s x_2 - 1) \left(\frac{\gamma \sigma_s x_3 y_3}{x_2} + \gamma K u_2(t_2 + F[t_1^*](t_2)) \right),$$

concluding the desired result from the fact that $\sigma_s x_2 < 1$ and $u_2(t_2 + F[t_1^*](t_2)) > 0$ (see (56)). $\square$

Next, we give the proofs of Lemmas 4.2 and 4.3 given in Section 4.

Proof of Lemma 4.2. For $t_2 \in [t_1^*, t_c]$, from (31) and (38) we have that

$$w_b(t_2) = \int_{t_2}^{t_2 + F[t_1^*](t_2)} \frac{\sigma_f x[t_1^*, t_2, F[t_1^*](t_2)](r) - 1}{y[t_1^*, t_2, F[t_1^*](t_2)](r)} dr. \quad (68)$$

From Remark 4.2 $w_b(t_2) > 0$ for all $t_2 \in [t_1^*, s_0]$. Since t_1^* is fixed, given $t_2 \in [t_1^*, t_c]$, the duration of the strict quarantine is given by $\mu = F[t_1^*](t_2)$. For simplicity of notation in this proof we neglect the parameters t_1^* and $F[t_1^*](t_2)$ from solutions $x[t_1^*, t_2, F[t_1^*](t_2)]$ and $y[t_1^*, t_2, F[t_1^*](t_2)]$ and write $x[t_2]$, $y[t_2]$. Given $t_2 \in (s_0, t_c)$, we define the auxiliary functions for $s \in (t_2, t_2 + F[t_1^*](t_2))$

$$g[t_2](s) = \sigma_f x[t_2](s) y[t_2](s) + (\sigma_f x[t_2](s) - 1)(\sigma_s x[t_2](s) - 1), \quad (69a)$$

$$f[t_2](s) = \frac{\sigma_f x[t_2](s) - 1}{y[t_2](s)}, \quad (69b)$$

$$i[t_2](s) = \frac{x[t_2](s)}{y[t_2](s)} (\sigma_f x[t_2](s) + \sigma_f y[t_2](s) - 1) + \gamma g[t_2](s) \int_t^s \frac{x[t_2](r)}{y[t_2](r)} dr. \quad (69c)$$

By computing the derivatives for $s \in (t_2, t_2 + F[t_1^*](t_2))$ we obtain

$$\frac{d}{ds} g[t_2](s) = -\gamma \sigma_s^2 x[t_2](s) y[t_2](s) (\sigma_f x[t_2](s) + \sigma_f y[t_2](s) - 1), \quad (70a)$$

$$\frac{d}{ds} f[t_2](s) = -\gamma \frac{g[t_2](s)}{y[t_2](s)}, \quad (70b)$$

$$\begin{aligned} \frac{d}{ds} i[t_2](s) &= -\gamma \sigma_s x[t_2](s) \left(\gamma \sigma_s y[t_2](s) \int_t^s \frac{x[t_2](r)}{y[t_2](r)} dr + 1 \right) (\sigma_f x[t_2](s) \\ &\quad + \sigma_f y[t_2](s) - 1), \end{aligned} \quad (70c)$$

and from (62) and (56)–(57) with $i = 2$ we have that

$$\begin{aligned} w_b'(t_2) &= f[t_2](t_2 + F[t_1^*](t_2)) - ft_2 \\ &\quad + \gamma y_2(\sigma_s - \frac{1}{x_2}) \int_{t_2}^{t_2 + F[t_1^*](t_2)} \frac{x[t_2](s)}{y[t_2]^2(s)} (\sigma_f x[t_2](s) \\ &\quad + \sigma_f y[t_2](s) - 1) ds \\ &\quad + \gamma y_2(\sigma_s - \frac{1}{x_2}) \int_{t_2}^{t_2 + F[t_1^*](t_2)} \gamma \left(\int_t^s \frac{x[t_2](r)}{y[t_2](r)} dr \right) \frac{g[t_2](s)}{y[t_2](s)} ds \\ &= f[t_2](t_2 + F[t_1^*](t_2)) - ft_2 + \gamma y_2(\sigma_s - \frac{1}{x_2}) \\ &\quad \times \int_{t_2}^{t_2 + F[t_1^*](t_2)} \frac{i[t_2](s)}{y[t_2](s)} ds \end{aligned} \quad (71)$$

Note that if $g[t_2]$ is a positive function, then from (70b) $f[t_2]$ is decreasing. In addition, if $i[t_2]$ is also a positive function, we can conclude that w_b is decreasing on $[t_1^*, t_c]$. In what follows we will show that, in fact, $g[t_2](s) > 0$ and $i[t_2](s) > 0$ for all $s \in (t_2, t_2 + F[t_1^*](t_2))$.

Since both derivatives of $g[t_2]$ and $i[t_2]$ have the opposite sign of $(\sigma_f x[t_2](s) + \sigma_f y[t_2](s) - 1)$, we need to analyse the latter function. First, assume $\sigma_f x[t_2](s) + \sigma_f y[t_2](s) > 1$ for all $s \in (t_2, t_2 + F[t_1^*](t_2))$, then from (70a), $g[t_2]$ is a decreasing function. Moreover, from Remark 4.2 we have that $\sigma_f x[t_2](t_2 + F[t_1^*](t_2)) - 1 < 0$ for $t_2 \in (s_0, t_c)$ which combined with the inequality $\sigma_s x[t_2](t_2 + F[t_1^*](t_2)) - 1 < 0$ yields $g[t_2](s) > g[t_2](t_2 + F[t_1^*](t_2)) > 0$ for all $s \in (t_2, t_2 + F[t_1^*](t_2))$. Thus from (69c) $i[t_2](s) > 0$ for all $s \in (t_2, t_2 + F[t_1^*](t_2))$.

From the fact that $x + y$ is a decreasing function and $x(t_2) + y(t_2) > 1/\sigma_f$, if we assume that there exists $s_3 \in [t_2, t_2 + F[t_1^*](t_2)]$ such that $\sigma_f x[t_2](s_3) + \sigma_f y[t_2](s_3) = 1$, from (70a) and (70c) we have that $g[t_2]$ and $i[t_2]$ attains a global minimum at s_3 . Thus $g[t_2](s) \geq g[t_2](s_3) =$

$\sigma_f y[t_2](s_3) > 0$ and $i[t_2](s) \geq i[t_2](s_3) = \gamma g[t_2](s_3) \int_t^{s_3} \frac{x[t_2](r)}{y[t_2](r)} dr > 0$ for all $s \in [t_2, t_2 + F[t_1^*](t_2)]$ concluding also in this case that $g[t_2]$ and $i[t_2]$ are positive functions.

Finally, from Remark 4.2 we deduce that w_b changes sign at most once on $[t_1^*, t_c]$. $\square$

Proof of Lemma 4.3. For $t_2 \in [t_c, t_f]$, $y_3 = y[t_1^*, t_2, T - t_2](T)$, from Eq. (64) we have the identity

$$\left(w_b(t_2) - \frac{1}{\gamma K} \right) y_3 = (\sigma_f - \sigma_s) y_3 \int_{t_2}^T \frac{x[t_1^*, t_2, T - t_2](r)}{y[t_1^*, t_2, T - t_2](r)} dr - \frac{1}{\gamma}. \quad (72)$$

Using that for $r \in [t_2, T]$

$$\frac{\partial}{\partial t_2} \left(\frac{y[t_1^*, t_2, T - t_2](T)}{y[t_1^*, t_2, T - t_2](r)} \right) = \frac{y[t_1^*, t_2, T - t_2](T)}{y[t_1^*, t_2, T - t_2](r)} \gamma \sigma_s \int_r^T \frac{\partial}{\partial t_2} x[t_1^*, t_2, T - t_2](r) dr,$$

Eqs. (60), (62a), the definition of u_2 given in (51a) and its positivity (see (56)) we obtain that the right hand side of Eq. (72) is decreasing as a function of t_2 yielding the left hand side is also decreasing as a function of t_2 .

The second statement follows from the definition of t_f from where we have that $G(t_f) = 0$ or $x(r) < \frac{1}{\sigma_f}$ for all $r \in [t_f, t_f + G(t_f)]$ implying that $w_b(t_f) \leq 0$. Also, assume $w_b(t_1^*) \leq 0$, then from Lemma 4.2 $s_0 = t_1^*$ and w_b is strictly decreasing on (t_1^*, t_c) from where $w_b(t_2) < 0$ for all $t_2 \in (t_1^*, t_c]$. Moreover, using that $w_b(t_c) \leq 0$, the positivity of γK and $y[t_1^*, t_c, T - t_c](T)$ and the monotonicity of the function $(w_b(t_2) - \frac{1}{\gamma K}) y[t_1^*, t_2, T - t_2](T)$ with respect to t_2 in the interval $[t_c, t_f]$ we obtain that $y[t_1^*, t_2, T - t_2](T) (w_b(t_2) - \frac{1}{\gamma K}) < 0$ for all $t_2 \in [t_c, t_f]$. Thus, from (37), we conclude that $\tilde{J}(t_2)$ is decreasing on (t_1^*, t_f) yielding $t_2^* = t_1^*$ and $\mu^* = G(t_1^*)$ contradicting the existence of a boundary arc (Lemma A.2). $\square$

Data availability

No data was used for the research described in the article.

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